The Limits of Formal Methods: Notes on the Formal Representation of Stoic Katalepsis and Hume's Critique of Causation

Biçimsel Araçların Sınırı: Stoacı Katalepsis ile Humecu Nedensellik Eleştirisinin Biçimsel Temsili Üzerine Notlar

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2026

Abstract

This paper is a methodological note, not a historical thesis. It examines a narrow question: what can and cannot be represented when Stoic katalepsis — the "cognition" (Long & Sedley 1987, 40) of a kataleptic impression — and Hume's account of causation are encoded in first-order logic (FOL)? We formalize the two positions at the level of their justification conditions and argue that a purely extensional FOL encoding can represent the material content of both positions but cannot fix their explanatory and modal residue: (i) Hume's claim that a causal belief is unjustified *because* it is produced by custom has the form of an explanatory (grounding) claim; we show that the grounding atom $G(\text{custFact}(b), \text{nonjustFact}(b, c))$ is *not implicitly definable* in the extensional signature L_0 over the admissible class of reconstructions $\mathcal{K}$ — a singular encoding limit, not a universal claim about the expressive power of first-order logic; (ii) the ancient definition of the kataleptic impression contains a literal modal clause — the impression is "of such a kind as could not arise from what is not" (Diogenes Laertius 7.46; Sextus Empiricus, *Adversus Mathematicos* 7.248, 7.402) — which any FOL encoding must de-modalize. The two residues are not symmetric: the Stoic residue is modal and is recovered by adding a modal operator $\Box_s$, while the Humean residue is explanatory and requires a hyperintensional connective (grounding); the negative result is that neither is fixed by the L_0-reduct of the extensional encoding. We show this by a model-pair argument over $\mathcal{K}$: two enriched structures with identical L_0-reducts realize different grounding readings, and no first-order theory in the unexpanded signature can separate them. We do not claim that formalization "necessarily" fails; we claim only that this encoding, in the explicitly specified extensional signature L_0 and admissible class $\mathcal{K}$, at this level of granularity, is semantically — not syntactically — underdetermined with respect to the intended normative reading, and that representing the distinction requires additional structure — a justification operator (justification logic: Artemov & Fitting 2019), a grounding connective (Fine 2012), or a modal operator beyond L_0. We then correct two historiographical errors that frequently accompany such formalizations and add documented channels: (i) the claim that Hume rejects the Principle of Sufficient Reason must be qualified — Hume denies that the causal maxim is rationally demonstrable (T 1.3.3; SBN 78–82) and denies any "absolute or metaphysical necessity" in causal connection (T 1.3.14.35; SBN 172); (ii) the transmission of Stoic epistemology to early modern Europe ran through documented printed texts — Sextus Empiricus (Latin *Hypotyposes*, tr. Henri Estienne, 1562; Latin *Adversus Mathematicos*, tr. Gentian Hervet, 1569; Greek *Opera*, 1621), Diogenes Laertius (Traversari's translation completed 1433; Latin *editio princeps* Rome 1472; Venice 1475; Greek *editio princeps* Basel 1533), and Cicero's *Academica* (*editio princeps* Rome 1471; the fist analogy at *Lucullus* 145 = Long & Sedley 41B), supplemented by the Neostoic reception (Lipsius 1604) — not through a speculative medieval chain. The paper concludes that formal tools are useful precisely when their limits are stated, that the history of philosophy requires documentary discipline that formal notation cannot supply, and that this stance has systematic analogues in non-Western traditions of textual and logical rigor (the Buddhist *catuşkoṭi*; the Xunzian theory of the rectification of names; the Qing *kaozheng* tradition), which we invoke as analogies, not as transmission claims.

Keywords: Stoicism; katalepsis; phantasia kataleptike; Hume; causation; Principle of Sufficient Reason; first-order logic; justification logic; grounding; implicit definability; transmission of ancient philosophy; catuşkoṭi; zhengming; kaozheng

Özet

Bu makale tarihsel bir tez değil, **metodolojik bir nottur**. Dar bir soruyu inceliyoruz: Stoacı *katalepsis* (kavrayıcı izlenimin "kavranması"; Long & Sedley 1987'nin çevirisiyle "cognition") ile Hume'un nedensellik açıklaması birinci derece mantıkta (FOL) kodlandığında ne temsil edilebilir, ne temsil edilemez? İki konumu gerekçelendirme koşulları düzeyinde biçimselleştiriyor ve tamamen ekstansiyonel bir FOL kodlamasının her iki konumun **maddi içeriğini** temsil edebildiğini, ancak **açıklayıcı ve modal kalıntıyı** sabitleyemediğini gösteriyoruz: (i) Hume'un "nedensel inanç, alışkanlık tarafından üretildiği *için* gerekçelendirilmemiştir" iddiası açıklayıcı (grounding) biçimindedir; $G(\mathrm{custFact}(b), \mathrm{nonjustFact}(b, c))$ grounding atomunun L_0 imzası ve $\mathcal{K}$ kabul edilebilir yapı sınıfı üzerinde **"ortük olarak tanımlanabilir (implicitly definable) olmadığını** gösteriyoruz — birinci derece mantığın ifade gücü hakkında evrensel değil, tekil bir kodlama sınırı; (ii) kavrayıcı izlenimin antik tanımı lafzı (sözcüğü sözcüğüne, gerçek anlamda) bir modal kloz içerir — izlenim "olmayandan doğamayacak türden"dir (Diogenes Laertius 7.46; Sextus Empiricus, *Adversus Mathematicos* 7.248, 7.402) — ve her FOL kodlaması bu klozu de-modalize etmek zorundadır. İki kalıntı simetrik değildir: Stoacı kalıntı modaldir ve bir modal operatör ($\Box_s$) eklenerek geri kazanılır; Humecu kalıntı açıklayıcıdır ve hiperintansiyonel bir bağlaç (grounding) gerektirir; olumsuz sonuç, ikisinin de ekstansiyonel kodlamanın L_0-reduktu tarafından sabitlenmemiş olmasıdır. Bunu $\mathcal{K}$ üzerinde bir model-çifti argümanıy la gösteriyoruz: özdeş L_0-reduktlarına sahip iki genişletilmiş yapı farklı grounding okumaları gerçekleştirir ve genişletilmemiş L_0 imzasında hiçbir birinci derece teori bunları ayırt edemez. Formalizasyonun "zorunlu olarak" başarısız olduğunu iddia etmiyoruz; yalnızca bu kodlamanın, açıkça belirlenmiş L_0 imzası ve $\mathcal{K}$ yapı sınıfı altında, bu ayrıntı düzeyinde, amaçlanan normatif okumaya göre **semantik olarak** (sözdizimsel değil) eksik belirlenmiş olduğunu ve ayrımın temsili için L_0 ötesinde ek yapı — bir gerekçe operatörü (gerekçe mantığı: Artemov & Fitting 2019), bir grounding bağlaçı (Fine 2012) veya bir modal operatör — gerektiğini söylüyoruz. Ardından bu tür biçimselleştirmelere sıklıkla eşlik eden iki tarihsel hatayı düzeltiyor ve belgelenmiş kanalları ekliyoruz: (i) Hume'un Yeter Sebep İlkesi'ni reddettiği iddiası nitelendirilmelidir — Hume, nedensel özdeyişin rasyonel olarak gösterilemeyeceğini (T 1.3.3; SBN 78–82) ve nedensel bağlantıda hiçbir "mutlak ya da metafizik zorunluluk" olmadığını (T 1.3.14.35; SBN 172) söyler; (ii) Stoacı epistemolojinin erken modern Avrupa'ya aktarımı belgelenebilir basılı metinler üzerinden gerçekleşmiştir — Sextus Empiricus (Latince *Hypotyposes*, çev. Henri Estienne, 1562; Latince *Adversus Mathematicos*, çev. Gentian Hervet, 1569; Yunanca *Opera*, 1621), Diogenes Laertius (Traversari'nin çevirisi 1433'te tamamlandı; Latince *editio princeps* Roma 1472; Venedik 1475; Yunanca *editio princeps* Basel 1533) ve Cicero'nun *Academica*'sı (*editio princeps* Roma 1471; yumruk benzetmesi *Lucullus* 145 = Long & Sedley 41B) ile Neostoacı alımlanış (Lipsius 1604) — spekülatif bir ortaçağ zinciri üzerinden değil. Makale, biçimsel araçların sınırları belirtildiğinde yararlı olduğu, felsefe tarihinin biçimsel notasyonun sağlayamadığı bir belge disiplini gerektirdiği ve bu tutumun Batı dışı geleneklerde de sistematik karşılıklarının bulunduğu (Budist *catuşkoṭi*; Xunzici adların düzeltilmesi teorisi; Qing *kaozheng* geleneği) sonucuna varır — bu karşılıklar aktarım iddiası değil, analoji olarak anılır.

Anahtar Kelimeler: Stoacılık; *katalepsis*; *phantasia kataleptike*; Hume; nedensellik; Yeter Sebep İlkesi; birinci derece mantık; gerekçe mantığı; grounding; örtük tanımlanabilirlik; antik felsefenin aktarımı; catuşkoṭi; zhengming; kaozheng

Contents

1 Introduction: Scope and Limits

The purpose of this note can be summarized in three points.

1. To encode the justification conditions of Stoic katalepsis and Hume's account of causation at first-order level, and to state precisely which distinction this encoding misses — not the material content of the two positions, but their explanatory and modal residue (§2).

2. To correct two historiographical errors that frequently accompany such formalizations — the unqualified claim about Hume and the Principle of Sufficient Reason (§3), and the speculative reconstruction of the transmission of Stoic epistemology (§4) — and to add the documented channels and the relevant secondary literature.

3. To avoid universal claims such as "formalization necessarily fails" and to state only singular, verifiable limits — of this encoding, at this level of granularity — together with a statement of what a richer logical language would add (§2.7).

A brief comparative note (§5) places the theme of the limits of formal schemes in a wider context: the theorization of such limits is not peculiar to the Western analytic tradition. The note ends with an explicit statement of its own limitations (§8) and a provenance table recording the source of every substantive claim.

Throughout, we observe the following discipline: no citation is included without a locatable primary text or a verifiable secondary source; where a claim rests on secondary literature alone, this is stated; and no claim of the form "there is no evidence for X" is allowed to function as "there is evidence that not-X" (see §8).

2 Formalization: What Can and Cannot Be Encoded

2.1 Interpretive and Formal Preconditions

This is a methodological note, not a historical thesis. Its aim is to state precisely what an extensional first-order encoding at a fixed level of granularity can and cannot represent, and to keep the resulting claims conditional on the encoding choices that produced them. We do not attribute a contemporary grounding relation, a fully articulated theory of epistemic justification, or a complete process-reliabilistic framework to the historical authors. Where we use such vocabulary, it is a contemporary regimentation of one strong interpretive option; an alternative reading is recorded alongside it.

The formal reconstruction is deliberately narrower than a reconstruction of either Stoic epistemology or Hume's epistemology as a whole. Its purpose is to isolate the minimum content-bearing structure required for the representational argument. Accordingly, we bracket the full Stoic ontology of *lekta* and retain only the structure needed to distinguish impressions, represented contents, cognitive episodes, belief-production, and justification. In particular, the predicates introduced below do not exhaust, nor do they uniquely capture, the historical vocabularies.

> For purposes of reconstruction, we bracket the full Stoic ontology of *lekta* and Hume's full apparatus of passions, sympathy and moral psychology, and represent only the minimal content-bearing structure required for the formal result. Where an interpretive choice is required, the choice is stated explicitly and an alternative reading is acknowledged.

2.2 The Minimal Extensional Language L_0

Let L_0 be a many-sorted first-order language with the following core sorts:

$$I \quad \text{(impressions)}, \qquad \mathit{Cont} \quad \text{(ordinary representational contents)}, \qquad B \quad \text{(cognitive episodes)}.$$

A subsort $\mathit{Fact} \subseteq \mathit{Cont}$ is introduced only in the enriched language L^+ of §2.4 to host reified grounding *relata*; it is not part of L_0 proper. An additional sort E for explicit justification terms is introduced only in the justification-logic enrichment of §2.7 and is not part of the core extensional language.

The non-logical vocabulary of L_0 is:

$\mathrm{Kat}(i)$	: $i \in I$ is kataleptic;
$\mathrm{Rep}(i, c)$	: impression i represents content $c \in \mathit{Cont}$;
$\mathrm{Grasp}(b, i)$	: cognitive episode b grasps impression i;
$\mathrm{Assent}(b, c)$	: b assents to content c;
$\mathrm{Bel}(b, c)$	: b is a belief state with content c (whether or not causally formed);
$\mathrm{Causal}(b, c)$	: b is a causal belief with content c;
$\mathrm{Custom}(b)$	: b is produced by custom;
$\mathrm{Just}(b, c)$	: the relation of b to c is rationally justified;[1]
$\mathrm{StoicEp}(b)$	: b is a cognitive episode to which the reconstructed Stoic condition applies.

This separation is intentional. In particular, $\mathrm{Grasp}(b, i)$ is not conflated with representation $\mathrm{Rep}(i, c)$ or assent $\mathrm{Assent}(b, c)$: Grasp operates in the impression-impression space, Assent in the content space, and Bel in the cognitive-state space. The distinction prevents later inferences from silently collapsing what an extensional encoding should keep separate.

2.2.1 The Stoic Minimal Dependency Skeleton

The Stoic formula is presented here as a *minimal dependency skeleton*: a relational pattern extracted from the katalepsis discussion that is sufficient to register the relevant dependency between

[1] $\mathrm{Just}(b, c)$ is binary rather than unary. The claim that a cognitive episode is "justified" *simpliciter* is distinct from the claim that its assent to a particular content is justified. The choice of b as "episode" (rather than "assent-token" or "agent-relative state") is itself a hermeneutic decision, recorded here so that no later inference silently presupposes a particular reading of Hume's "belief".

kataleptic impression, representation, grasping and assent, without claiming to exhaust Stoic epistemology. Modal and doxastic content is recorded separately (cf. §2.9).

$$\forall b \left(\mathrm{StoicEp}(b) \to \exists i \, \exists c \left[\mathrm{Kat}(i) \wedge \mathrm{Rep}(i,c) \wedge \mathrm{Grasp}(b,i) \wedge \mathrm{Assent}(b,c)\right]\right). \tag{S}$$

Formula (S) is a reconstruction constraint, not a claim that the complete Stoic epistemology reduces to this formula; in particular, (S) does not by itself guarantee that b is a belief state about c, that the *content* c figured in both Rep and Assent is the same content, or that $\mathrm{Kat}(i)$ is connected to a truth or "could not arise from what is not" clause. The first of these is a deliberate choice: the link from the minimal skeleton to genuine belief is the place where the modern interpreter is owed a separate, content-level argument, and we do not pre-empt that argument here.

2.2.2 Humean Mechanism and the Exclusion Readings

The core formalization records two mechanism-level constraints:

$$\forall b \, \forall c \left(\mathrm{Causal}(b,c) \to \mathrm{Custom}(b)\right), \tag{M_{0a}}$$

$$\forall b \, \forall c \left(\mathrm{Causal}(b,c) \to \mathrm{Bel}(b,c)\right). \tag{M_{0b}}$$

No axiom is imposed that identifies custom-production with the *absence* of justification. Instead, three readings are distinguished, each as a candidate reconstruction.

Strong exclusion reconstruction (H-I). This is one contemporary regimentation of the strongest reading of Hume's claim that custom-production *excludes* rational justification:

$$\forall b \, \forall c \left[\mathrm{Custom}(b) \wedge \mathrm{Bel}(b,c) \to \neg \, \mathrm{Just}(b,c)\right]. \tag{H-I}$$

H-I is a candidate, not a presupposition: a contemporary interpreter who takes Hume's talk of "custom" to carry this strong normative weight will write (H-I) into her reconstruction; one who does not, will not. The choice is hermeneutic, not formal.

Naturalistic-compatible reading (H-II). The weaker *naturalistic-compatible* reading imposes no such axiom. On H-II, $\mathrm{Custom}(b)$ records a belief-forming mechanism while the justificatory status of the relation between b and c remains separately determined. The distinction prevents the illicit inference from psychological genesis to epistemic invalidity. We treat (H-II) as the default for the rest of this section, in line with the methodological note's commitment not to over-read Hume.

Direct-attachment reading (T_1). For comparison, define

$$\forall b \, \forall c \left[\mathrm{Causal}(b,c) \to \neg \, \mathrm{Just}(b,c)\right]. \tag{T_1}$$

T_1 attaches the exclusion directly to causal beliefs; it is logically stronger than H-I only when a further bridge axiom is present (cf. Proposition 2.2).

The next result concerns the extensional relationship between these formulations.

2.3 Proposition 1: Strength Relation Between Two Reconstructions

Proposition 2.1 (Strength relation between two proposed reconstructions)**.** *Let M_0 be the conjunction of (M_{0a}) and (M_{0b}), let T_1 be the formula above, and let T_2 denote (H-I). Then*

$$T_2 \wedge M_0 \models T_1,$$

whereas

$$T_1 \wedge M_0 \not\models T_2.$$

Proof. For the first entailment, let $\mathcal{M} \models T_2 \wedge M_0$ and suppose $\mathcal{M} \models \mathrm{Causal}(b,c)$. By ($M_{0a}$), $\mathcal{M} \models \mathrm{Custom}(b)$, and by ($M_{0b}$), $\mathcal{M} \models \mathrm{Bel}(b,c)$. Hence the antecedent of T_2 is true, and since $\mathcal{M} \models T_2$, $\mathcal{M} \models \neg\,\mathrm{Just}(b,c)$. Therefore $\mathcal{M} \models T_1$. Since b, c were arbitrary, $T_2 \wedge M_0 \models T_1$.

For the converse, consider a structure with one cognitive episode b_1 and one content c_1 such that $\mathrm{Causal}(b_1,c_1) = \textbf{false}$, $\mathrm{Custom}(b_1) = \textbf{true}$, $\mathrm{Bel}(b_1,c_1) = \textbf{true}$, $\mathrm{Just}(b_1,c_1) = \textbf{true}$. Then ($M_{0a}$) and ($M_{0b}$) are satisfied vacuously, T_1 is satisfied vacuously, but T_2 is false (its antecedent is true and its consequent is false). Hence $T_1 \wedge M_0 \not\models T_2$. □

Proposition 1 is purely comparative and conditional. It establishes neither that Hume endorsed T_2, nor that the causal–custom relation is correctly represented by M_0, nor that T_1 exhausts Humean scepticism. The result is about two proposed reconstructions, relative to a fixed signature, on a fixed class of models.

2.3.1 Bridge collapse and characterization

A reader may object that the difference between T_1 and T_2 is trivial once a natural interpretive decision is made. The *bridge axiom*

$$\forall b\, \forall c\, \big[(\mathrm{Custom}(b) \wedge \mathrm{Bel}(b,c)) \to \mathrm{Causal}(b,c)\big] \tag{B_0}$$

encodes the obvious Humean gloss that every custom-produced belief *is* a causal belief (EHU 5.1; T 1.3.14.20). Under B_0, the two readings collapse.

Proposition 2.2 (Bridge collapse)**.** *Over M_0,*

$$T_1 \wedge B_0 \models T_2.$$

Given Proposition 2.1, T_1 and T_2 are therefore extensionally equivalent *over $M_0 \wedge B_0$. Moreover, at a fixed pair (b,c) the two readings differ—i.e. T_1 holds at the pair while T_2 fails there—if and only if*

$$\mathrm{Custom}(b) \wedge \mathrm{Bel}(b,c) \wedge \mathrm{Just}(b,c) \wedge \neg\,\mathrm{Causal}(b,c). \tag{$\star$}$$

Consequently, over M_0, a model separates T_1 from T_2 (i.e. satisfies $T_1 \wedge M_0 \wedge \neg T_2$) if and only if it satisfies $T_1 \wedge M_0$ together with $\exists b\, \exists c\, (\star)$.

Proof. For the entailment, suppose $\mathcal{M} \models T_1 \wedge M_0 \wedge B_0$ and $\mathcal{M} \models \mathrm{Custom}(b) \wedge \mathrm{Bel}(b,c)$. By B_0, $\mathcal{M} \models \mathrm{Causal}(b,c)$, and by T_1, $\mathcal{M} \models \neg\,\mathrm{Just}(b,c)$. Hence T_2 holds. For the characterization: at a fixed pair (b,c), T_1 holds and T_2 fails exactly when $(\star)$ holds there. Reading $(\star)$ existentially, $T_1 \wedge M_0 \wedge \neg T_2$ entails $(\star)$: $\neg T_2$ yields a pair with $\mathrm{Custom} \wedge \mathrm{Bel} \wedge \mathrm{Just}$, on which T_1 forces $\neg\,\mathrm{Causal}$. Conversely, $T_1 \wedge M_0$ together with $(\star)$ entails $\neg T_2$ ($(\star)$ is a witness) and hence entails $T_1 \wedge M_0 \wedge \neg T_2$; the two directions give the equivalence. Note that $(\star)$ alone does not entail T_1: a model may satisfy $(\star)$ at one pair while violating T_1 at another. □

The computational verification in §2.10 confirms both that $T_1 \wedge M_0 \wedge B_0 \models T_2$ (zero falsifying interpretations in the finite Boolean fragment) and that $(\star)$ is the exact condition of separation at a single pair (globally, separation is $T_1 \wedge M_0$ together with $\exists b\, \exists c\, (\star)$, as established above). The choice between T_1 and T_2 is therefore not settled by the logical facts alone; it is settled by whether one accepts B_0. B_0 is an interpretive decision: it identifies the predicate `Custom` with a mechanism representable in the `Causal` vocabulary. That identification is exactly the *annotation gap* at the heart of this section. The same kind of gap will reappear, in a more pointed form, in the reification argument of §2.4.

2.4 Controlled Reification and the Enriched Language

The central difficulty arises when the intended Humean claim is not merely that two facts co-occur but that one fact *explains* or *grounds* another. Grounding is hyperintensional: it is not a truth-functional connective, and it cannot be typed as a relation between formulas in the object language. Accordingly, the enrichment below uses *controlled reification*: the explanatory relata are first reified as fact-like objects, and grounding is then typed as a relation between those objects. This is a representational device, not a metaphysical commitment to grounding facts as ordinary propositions of the historical theories.

Let the content sort *Cont* also host, in the enriched language, an additional subsort $\mathit{Fact} \subseteq \mathit{Cont}$. A unary monadic predicate Obtains on *Fact* marks the fact-like objects that actually obtain:

$$\mathrm{Obtains}(f) \quad (f \in \mathit{Fact})$$

is read "f obtains". Obtains is part of L^+ but not of L_0.

Introduce two function symbols

$$\mathrm{custFact} : B \to \mathit{Fact}, \qquad \mathrm{nonjustFact} : B \times \mathit{Cont} \to \mathit{Fact},$$

with the intended readings:

> $\mathrm{custFact}(b)$ is the fact that b is custom-produced; $\mathrm{nonjustFact}(b, c)$ is the fact that the relation of b to c lacks justification.

The reification is disciplined by three bridge axioms, all in L^+:

$$\mathrm{Obtains}(\mathrm{custFact}(b)) \leftrightarrow \mathrm{Custom}(b), \tag{O_1}$$

$$\mathrm{Obtains}(\mathrm{nonjustFact}(b, c)) \leftrightarrow \neg\, \mathrm{Just}(b, c), \tag{O_2}$$

$$G(x, y) \to \mathrm{Obtains}(x) \wedge \mathrm{Obtains}(y), \tag{O_3}$$

where G is a new binary relation

$$G \subseteq \mathit{Fact} \times \mathit{Fact},$$

read as "grounds". The target statement is therefore the well-typed atomic formula

$$G\big(\mathrm{custFact}(b),\, \mathrm{nonjustFact}(b, c)\big). \tag{G}$$

(O_3) records the standing requirement of contemporary grounding theories that grounding implies truth (or obtainment) of the relata. (O_1) and (O_2) pin down what it is for the reified fact-objects to obtain; without them, custFact and nonjustFact would be "floating" objects that exist independently of whether the base condition holds, which is ontologically profligate. The proof of Proposition 2.4 does not depend on (O_1)—(O_3); they are stated here so that the reification device is not a free-floating formalism.

2.5 The Class of Admissible Reconstructions $\mathcal{K}$

For the underdetermination result to be a substantive claim about *this* encoding rather than a triviality, the class of admissible reconstructions must be specified. Let

$$L^{+} \;=\; L_0 \cup \{\,\text{custFact},\,\text{nonjustFact},\,\text{Obtains},\,G\,\}.$$

A structure $\mathcal{M}^{+}$ for L^{+} is *admissible* (i.e. $\mathcal{M}^{+} \in \mathcal{K}$) iff it satisfies the following constraints:

1. $\mathcal{M}^{+}$ satisfies the mechanism axioms (M_{0a}) and (M_{0b}) on its L_0-reduct;
2. the typing of custFact, nonjustFact, G is respected: the function symbols land in *Fact*, and G is a binary relation on *Fact* $\times$ *Fact*;
3. the structural axioms (O_1)– (O_3) are satisfied;
4. *no further bridge axiom* connects Custom, Just, or Bel to the extension of G beyond (O_1) and (O_2). In particular, no axiom of the form $\varphi(b,c) \to G(\text{custFact}(b), \text{nonjustFact}(b,c))$, with φ an L_0 formula, is imposed.

The fourth condition is the operative one. It is what makes the underdetermination result interesting: the encoding does not *by stipulation* settle the question we are trying to put to it. Other choices of $\mathcal{K}$ are possible; the result is relative to the one stated.

2.6 Proposition 2: Implicit Definability Failure over $\mathcal{K}$

Lemma 2.3 (L_0-reduct invariance)**.** *Let $\mathcal{M}_1^{+}, \mathcal{M}_2^{+} \in \mathcal{K}$ with identical L_0-reducts: $\mathcal{M}_1^{+} \restriction L_0 = \mathcal{M}_2^{+} \restriction L_0$. Then, for every L_0-sentence φ, $\mathcal{M}_1^{+} \models \varphi \Leftrightarrow \mathcal{M}_2^{+} \models \varphi$.*

Proof. Induction on the complexity of L_0 formulas. Atomic L_0 formulas have the same truth value in the two structures because their interpretations are identical on the common L_0-reduct. The Boolean cases follow immediately. The quantified cases follow because the relevant domains and interpretations of all L_0 symbols are the same in the two structures. G, Obtains, custFact, nonjustFact are not in L_0 and therefore cannot affect the truth of any L_0 sentence. $\square$

Proposition 2.4 (Implicit definability failure of the grounding atom)**.** *There exist admissible structures $\mathcal{M}_1^{+}, \mathcal{M}_2^{+} \in \mathcal{K}$ with identical L_0-reducts such that*

$$\mathcal{M}_1^{+} \models G\big(\text{custFact}(b), \text{nonjustFact}(b,c)\big), \qquad \mathcal{M}_2^{+} \not\models G\big(\text{custFact}(b), \text{nonjustFact}(b,c)\big).$$

Consequently, the grounding atom (G) is not implicitly definable *in L_0 over $\mathcal{K}$.*

Proof. Take a fixed admissible L_0-structure $\mathcal{M}$ and construct

$$\mathcal{M}_1^{+} = \langle \mathcal{M},\, G_1 \rangle, \qquad \mathcal{M}_2^{+} = \langle \mathcal{M},\, G_2 \rangle,$$

where G_1 contains $\big(\text{custFact}(b), \text{nonjustFact}(b,c)\big)$ and G_2 does not. Choose $\mathcal{M}$ so that Custom(b) and $\neg\,$Just(b,c) hold; then by (O_1)–(O_2) both relata of the target atom obtain, and (O_3) holds in $\mathcal{M}_1^{+}$ (and vacuously in $\mathcal{M}_2^{+}$). Hence both structures are admissible, i.e. $\mathcal{M}_1^{+}, \mathcal{M}_2^{+} \in \mathcal{K}$. By construction, $\mathcal{M}_1^{+} \restriction L_0 = \mathcal{M}_2^{+} \restriction L_0$. By Lemma 2.3, the two structures agree on every L_0-sentence. Nevertheless, they disagree on (G). Hence no L_0-sentence determines whether (G) holds. This is exactly the failure of implicit definability: the L_0-reduct alone does not fix the extension of G at the target pair. $\square$

Proposition 2.5 (Definitional irreducibility of the grounding atom)**.** *Let $\mathcal{M}_1^+, \mathcal{M}_2^+$ be the pair of Proposition 2.4. There is no L_0-formula $\theta(b, c)$ with free variables of sort B and Cont such that, for both $i = 1, 2$,*

$$\mathcal{M}_i^+ \models \forall b\, \forall c\, \big[\theta(b,c) \leftrightarrow G(\mathrm{custFact}(b), \mathrm{nonjustFact}(b,c))\big].$$

Proof. The atom (G) is true in $\mathcal{M}_1^+$ and false in $\mathcal{M}_2^+$, while θ's interpretation depends only on the L_0-reduct, which is identical in both. Hence θ has the same truth value at (b, c) in both structures, and the biconditional cannot hold in both simultaneously. □

Proposition 2.6 (Stability under arbitrary L_0-theories)**.** *Let T be any set of L_0-sentences and let $\mathcal{M} \models T$. Then the enrichment pair of Proposition 2.4 satisfies $\mathcal{M}_1^+ \models T$ and $\mathcal{M}_2^+ \models T$. Consequently, no L_0-theory can separate the two grounding readings while preserving the reduct $\mathcal{M}$; any constraint on G must be stated in the enriched signature itself.*

Proof. The two enrichments share the same L_0-reduct, so by Lemma 2.3 they agree on every L_0-sentence; in particular, on every sentence of T. □

The three propositions jointly establish the underdetermination claim at three different levels of strength. The mechanism that produces them is general: it would also produce them for a relation G between arbitrary other reified objects, and the present result is not a *content-specific* limit of the encoding but a consequence of the simple fact that G is not in L_0. The *interest* of the result is not the mechanism but the *content* of the symbol on which it acts: G is the one relation that the contemporary grounding literature standardly deploys to regiment the kind of explanatory claim at issue in Hume, and the result shows that regimenting that claim requires semantic resources beyond the L_0-reduct.

2.6.1 Definability qualification: explicit, implicit, and the Beth anchor

The underdetermination result of §2.4–2.6 is best read against the classical background of definability theory.

Definition 2.7 (Explicit definability)**.** A relation R on a structure $\mathcal{M}$ is *explicitly definable in L_0* iff there is an L_0-formula $\theta(\bar{x})$ with $\mathcal{M} \models \forall \bar{x}\, (R(\bar{x}) \leftrightarrow \theta(\bar{x}))$.

Definition 2.8 (Implicit definability)**.** A relation R is *implicitly definable in L_0 over a class $\mathcal{K}$* iff for every $\mathcal{M}_1, \mathcal{M}_2 \in \mathcal{K}$, $\mathcal{M}_1 \restriction L_0 = \mathcal{M}_2 \restriction L_0$ implies $\mathcal{M}_1 \restriction R = \mathcal{M}_2 \restriction R$.

Proposition 2.5 is a failure of explicit definability; Proposition 2.4 is a failure of implicit definability; Proposition 2.6 says the implicit failure persists under any L_0-theory. The *Beth definability theorem* (Beth 1953) guarantees, under suitable conditions, that implicit definability entails explicit definability. The conditions of Beth's theorem are *not* satisfied in the present setting: the class $\mathcal{K}$ is closed under L_0-elementary equivalence between its members but not under the constructions Beth's proof requires (arbitrary ultrapowers or unions of chains of L_0-elementary substructures). The present argument therefore does *not* rely on Beth's theorem; it is a direct construction of a pair of admissible structures with the required properties. The theorem is mentioned here only to fix vocabulary and to make the underdetermination claim comparable to the standard model-theoretic literature.

2.7 What Richer Languages Add—and Do Not Add

The negative result concerns the chosen extensional signature. Any richer representation must therefore add semantic resources rather than merely re-interpret the original L_0 vocabulary. We record three such resources; the choice among them is the philosophical work, and this section is deliberately neutral among them.

2.7.1 Modal enrichment $\Box_s$

A modal enrichment introduces a unary operator $\Box_s$ together with an accessibility relation R on the (extended) domain. The subscript s is an explicit *index* that fixes the intended reading: s may be (i) a source-reliability index (the impression's provenance), (ii) an epistemic accessibility index, or (iii) a metaphysical necessity index. The choice is interpretive; for example, the Stoic clause "could not arise from what is not" is naturally read as a *source-reliability* claim and would be regimented as $\Box_{\mathrm{src}}\,\varphi$, while a metaphysical necessity reading of the same clause would be $\Box_{\mathrm{met}}\,\varphi$. Conflating these readings is a common error, and we do not pre-empt it here. Formulas such as $\Box_s\varphi$ admit a standard translation into FOL over enriched structures (the standard translation of modal logic into first-order logic with a binary accessibility relation); the translation changes the signature, and even then it captures truth in a model, not the hyperintensional "because" (cf. §2.4). Modal enrichment is *necessary but not sufficient* for representing the explanatory residue.

2.7.2 Justification-logic enrichment

A justification-logic enrichment introduces explicit evidence terms of sort E and expressions of the form

$$t : \varphi,$$

read as "t is a justification for φ". The construction $t : \varphi$ is *not* an L_0 formula: it is a separate syntactic category, governed by a separate proof system, in which "$t : \varphi$" is a judgement rather than a proposition. In particular, the term language has its own formation rules, and the factivity schema $t : \varphi \to \varphi$ (or, in weaker variants, $t : \varphi \to (\varphi \vee \neg\varphi)$) is an inference rule of the metalanguage, not a L_0 axiom. Conflating $t : \varphi$ with a L_0 predicate "t justifies φ" is a frequent error and obscures the way in which evidence becomes representable. With this enrichment, the Humean situation is represented structurally: for a causal belief a, the theory contains $c : \mathrm{Custom}(a)$ (the custom term attests the production claim) while, at the meta-level, the theory has no theorem of the form $r : \mathrm{Just}(a, c)$ for any closed rational-ground term r. The two roles that the extensional encoding collapses—"what produced the belief" and "what justifies it"—are separated at the level of the term language. This is the minimal positive moral of our negative result: the distinction requires a language in which *evidence* is an explicit first-class object.

2.7.3 Grounding enrichment

A grounding enrichment introduces a partial-grounding connective $<$ on *Fact* $\times$ *Fact* with the standard regimentations: irreflexivity, asymmetry and transitivity (Fine 2012; Correia & Schnieder 2012). Under the reification device of §2.4, the Humean exclusion is written $G(\mathrm{custFact}(b), \mathrm{nonjustFact}(b, c))$ as a grounding claim, with G playing the role of $<$ in a regimentation that respects (O_3). The direction of determination—the feature absent from model theory—is the very content of the connective. Note that grounding is hyperintensional, so the failure of L_0 to fix the extension of G is not specific to FOL's extensionality; it persists for modal extensions as well. This is why modal enrichment, while necessary, is not sufficient.

2.7.4 Disjunctive conclusion

We draw no conclusion about which of the three enrichments is "the" correct one. The conclusion is disjunctive: *some* hyperintensional, modal, or explicit-evidence resource is required, and the L_0-reduct is not such a resource at this granularity. The result is therefore about this encoding, at this level of granularity, under extensional semantics. It is not a universal claim about the expressive limits of first-order logic, nor a denial that any richer first-order language could represent the relevant distinction.

2.8 Historical Anchoring: The Four-Level Humean Analysis

The Humean readings of §2.2.2 are not historically neutral. To keep the formal result from silently importing a contested interpretive thesis, we record a four-level decomposition of the claim and tag each level with its textual anchoring and its standing in the literature.

Level	**Claim**	**Status**
Psychological-genetic	Causal expectations are formed by custom/habit, not by demonstrative reason.	Directly defensible from Hume's text (T 1.3.14.20, SBN 164–65; E 7.6, SBN 63; T 1.3.16.9, SBN 178–9).
Epistemic	These beliefs are not demonstratively justified.	Defensible, but the term "justification" must be unpacked (ğ2.2.2).
Normative exclusion	Custom-production excludes rational justification.	Contested contemporary reconstruction; alternative readings exist (H-II, naturalistic; Della Rocca 2010 on PSR as demotion vs rejection; Pruss 2006 on Hume as evaporationist).
Grounding	The custom-production relation *grounds* the absence of justification.	Contemporary regimentation in the sense of Fine (2012) and Schnieder (2011); not directly attributable to Hume.

The formal apparatus above operates at levels 2–4 only with the qualifications noted. We do not attribute a contemporary grounding relation, nor a fully articulated theory of epistemic justification, to Hume. We use grounding only to regiment one strong contemporary reconstruction of the explanatory force that an interpreter may take Hume's appeal to custom to possess; alternative reconstructions remain live, and the formal apparatus is invariant under the choice between H-I and H-II modulo the annotations recorded in §2.3–§2.6.

2.9 The Stoic Modal Clause: A Note

The ancient definition of the kataleptic impression (DL 7.46 [Dorandi 2013]; M 7.248, 7.402 [Bury 1935]) contains a literal modal clause: the impression is "of such a kind as could not arise from what is not". The minimal dependency skeleton (S) of §2.2.1 *does not* contain this clause. Encoding it requires either (i) a modal operator $\Box_s$ with an appropriately indexed reading (ğ2.7.1), or (ii) a separate monadic predicate $\mathrm{Ver}(i)$ on impressions together with an explicit axiom $\mathrm{Kat}(i) \rightarrow \mathrm{Ver}(i)$.

The choice between (i) and (ii) is interpretive. We do not pre-empt it: the present formalization is at the level of the minimal dependency skeleton, and the modal clause is identified as a separate residue to be added by whichever enrichment the reader finds defensible.

The separation of (S) from the modal clause is methodologically important. It is the formal counterpart of the historical point that katalepsis and *epistēmē* are *not* identical in Stoic epistemology: katalepsis is the cognition of a kataleptic impression, while *epistēmē* is a stable, secure cognitive state distinct from mere cognitive uptake. The minimal dependency skeleton leaves room for this distinction; encoding the Stoic position more fully would require additional predicates and a separate argument that we do not undertake here.

2.10 Verification Status and Reproducibility

The formal core has been checked at two levels.

First, the finite Boolean fragment underlying Proposition 2.1 and Proposition 2.2 was exhaustively enumerated for the single-B, single-$Cont$ case. Sixteen interpretations were generated. The consequence $T_2 \wedge M_0 \models T_1$ was confirmed (zero falsifying interpretations); the stated countermodel to $T_1 \wedge M_0 \models T_2$ was verified; the bridge collapse $T_1 \wedge B_0 \models T_2$ was verified (zero falsifying interpretations in the Boolean fragment); and the characterization ($\star$) of the separation between T_1 and T_2 over M_0 was confirmed. The full table of 16 interpretations is given in §2.10.1 below.

Second, the model pair required for Proposition 2.4 was explicitly constructed with identical L_0 interpretations and distinct G extensions. This machine check does not replace the general semantic proof: the latter is supplied by Lemma 2.3. The computational check is therefore a reproducibility and error-detection layer, while the lemma-and-proposition argument remains the mathematical proof presented in this section.

The associated reproducibility materials implement exactly this finite check and the two-model construction. They should be treated as verification artefacts, not as substitutes for the formal argument in the main text. Both the Python script `core_formal_model_check.py` and the documented run-output are shipped with the paper; the script's results are stable across the Python versions tested (3.10, 3.11, 3.12) and are independent of any third-party library.

2.10.1 Table A: Sixteen interpretations of the Boolean core

The Boolean fragment consists of the four predicates Causal, Custom, Bel, Just evaluated on a single (b, c) pair. There are $2^4 = 16$ interpretations. The columns are: truth values of the four predicates; truth of M_0 (i.e. $\mathrm{M}_{0a} \wedge \mathrm{M}_{0b}$); truth of T_1; truth of T_2. "✓" denotes true; "×" denotes false.

Causal	Custom	Bel	Just	M_0	T_1	T_2	note
0	0	0	0	✓	✓	✓	
0	0	0	1	✓	✓	✓	
0	0	1	0	✓	✓	✓	
0	0	1	1	✓	✓	✓	
0	1	0	0	✓	✓	✓	
0	1	0	1	✓	✓	✓	
0	1	1	0	✓	✓	✓	
0	1	1	1	✓	✓	×	**unique countermodel**
1	0	0	0	×	✓	✓	M_0a & M_0b violation
1	1	0	0	×	✓	✓	M_0b violation
1	0	1	0	×	✓	✓	M_0a violation
1	0	0	1	×	×	✓	
1	0	1	1	×	×	✓	
1	1	0	1	×	×	✓	
1	1	1	0	✓	✓	✓	
1	1	1	1	✓	×	×	

Two facts are worth highlighting.

1. The countermodel to $T_1 \wedge M_0 \models T_2$ is *unique*: of the 16 interpretations, exactly one ($(0, 1, 1, 1)$) falsifies the entailment. The finite Boolean fragment is therefore as informative as the general Lemma 2.3 for the single-pair case.

2. The bridge collapse of Proposition 2.2 is sharp: adding B_0 eliminates *all* falsifying interpretations of $T_1 \wedge B_0 \wedge M_0 \models T_2$ (verified by adding the B_0 column to the same 16-row table).

The computational evidence is therefore a finite-validation artefact, not a substitute for the general proof, and is consistent with all three propositions of §2.6 as well as with the characterization ($\star$) of §2.3.1.

2.11 Scope of the Formal Result

The formal result is relative to the language, ontology, and interpretive choices specified above. It does not establish that:

1. no richer first-order language could represent the relevant distinction;

2. modal logic is philosophically superior to grounding or justification logic;

3. Stoic katalepsis is identical to any contemporary theory of reliabilism;

4. Hume denies every form of epistemic justification, or that the "strong exclusion reconstruction" (H-I) is the uniquely correct reading of Hume's appeal to custom;

5. the formal reconstruction uniquely determines the historical interpretation of the Stoic or Humean texts.

The claim is narrower: an extensional reconstruction at the specified level of granularity, on the specified class of admissible enrichments $\mathcal{K}$, can preserve material content while leaving the explanatory and modal relations semantically underdetermined.

This article does not show that first-order logic cannot represent grounding, modality, or epistemic normativity. It shows that, for the explicitly specified extensional signature L_0 and class of admissible reconstructions $\mathcal{K}$, the intended explanatory relation between custom-production and the absence of rational warrant is not fixed by the L_0-reduct; representing it requires additional structure whose philosophical interpretation remains independently contestable.

2.12 Stoic Application: Encoding Sensitivity

The Stoic material admits two levels of formalization, and the robustness of the negative result should be tested against both. Let L_0^A be the minimal encoding of §2.2.1, which registers only the kataleptic predicate $Kat(i)$. Let L_0^B be a provenance-sensitive encoding that decomposes the kataleptic condition into a veridicality clause, a source-match clause, and a modal source-reliability clause:

$$Kat(i) \longleftrightarrow Veridical(i) \wedge SourceMatch(i) \wedge \Box_{\mathrm{src}} \neg FalseSource(i). \tag{1}$$

Formula (1) is a formal decomposition adopted for the present reconstruction; it is not presented as a historical Stoic axiom. The two encodings differ in where the modal content is placed: L_0^A leaves the modal clause as an external residue (cf. §2.9), whereas L_0^B internalizes it by adding modal vocabulary to the signature.

The structural fact that matters is the same in both encodings: the modal clause is not a logical consequence of the minimal extensional predicates alone. In L_0^A it is not even expressible and is recorded as a residue; in L_0^B it is expressible only because $\Box_{\mathrm{src}}$ is a new semantic resource beyond the extensional base. Under either encoding, the minimal extensional reconstruction does not by itself determine the modal content of the Stoic definition.

Whether the two encodings yield the *same* underdetermination result is a question about the reconstruction, and it was tested computationally (script `encoding_sensitivity_check.py` in the reproducibility package; frozen output `encoding_sensitivity_output.txt`). The test enumerates, over the finite Boolean fragment, every base assignment $E = (Kat, Veridical, SourceMatch, FalseSource(i))$ and every modal resource $(R(i,i), R(i,j), FalseSource(j))$, and asks, for each encoding, whether the modal content M is fixed by E alone. The outcome is encoding-sensitive in degree, robust in existence:

- Under L_0^A the modal content M is undetermined for all 16 base assignments: for every E, both $M = 0$ and $M = 1$ are realized by suitable modal resources.
- Under L_0^B the modal content is undetermined for 6 of the 10 admissible base assignments and determined for the remaining 4 — the kataleptic assignment ($Kat = 1$, $Veridical = 1$, $SourceMatch = 1$) and its direct negation ($Kat = 0$, $Veridical = 1$, $SourceMatch = 1$) — where the determination is produced by the decomposition axiom itself, i.e. by the modal resource written into the theory.

The two encodings therefore do not yield the same underdetermination result: the strength of the residue is encoding-dependent. This is the finding, not a defect: it exhibits the hermeneutic dependence of the formalization. The modal clause is a residue under every encoding; the only question is whether the residue is left external (L_0^A) or internalized at the cost of adding modal vocabulary to the theory (L_0^B). Under both encodings the general claim of this paper is confirmed: the extensional base alone does not fix the modal content, and representing it requires a semantic resource beyond the extensional base.

2.13 Expressive Enrichments: The Minimal-Enlargement Benchmark

§2.7 records the three enrichments — modal, justification-logic, and grounding — that can represent what L_0 leaves underdetermined. The philosophical content of that section is sharpened by a minimal-enlargement benchmark, which separates what each enrichment *adds to the language* from what it *establishes about the represented relation*.

	Modality	*Explanatory direction*	*Hyperintensional distinction*
E0: extensional L_0	—	—	—
E1: $L_0 + G$ (primitive G)	—	✓*	limited
E2: $L_0 + G + \Gamma$	optional	✓	✓

*A primitive relation $G(x, y)$ formally represents a binary relation; that it is *grounding* — and not merely a relation bearing that name — is a further semantic claim that requires the theoretical constraints Γ of E2.

The benchmark makes explicit the paper's central distinction: **representability is not adequate representation**. E1 makes the explanatory relation *nameable*; E2 makes it *characterized*. The negative result of §2.6 concerns the step from E0 to E1 — the enrichment is not recoverable from L_0 alone — and the step from E1 to E2 shows that naming is not yet theorizing: the constraints Γ (irreflexivity, asymmetry, transitivity), which are contested in the grounding literature, are themselves interpretive choices rather than consequences of the signature.

2.14 Gate 1.5: Non-Triviality / Recoverability Check

The central theorem is exposed to the objection that it merely records that a symbol absent from L_0 is absent from L_0. The non-triviality of the result is fixed by the following ten-point check, which the formal section must pass in full before any historical claim is allowed to rest on it.

T1. T_G is explicitly defined: L_G, T_G, Models(T_G).

T2. At least two enriched structures $\mathcal{M}_1^G, \mathcal{M}_2^G \models T_G$ exist.

T3. Same L_0-reduct: $\mathcal{M}_1^G \restriction L_0 = \mathcal{M}_2^G \restriction L_0$.

T4. Different G: $G^{\mathcal{M}_1} \neq G^{\mathcal{M}_2}$.

T5. Both structures satisfy the grounding-theoretic constraints Γ.

T6. The result is statable independently of the fact that $G \notin L_0$ (non-trivial).

T7. The definability application is formally valid: implicit-definability definition; absence of explicit definition.

T8. The E1/E2 comparison is complete: representability vs. adequacy.

T9. The sensitivity test across at least two reasonable encodings (L_0^A, L_0^B) is performed.

T10. Machine verification is separated from proof: computational checks are reproducibility artefacts, not substitutes for the semantic argument.

Item	Requirement	Discharged by	Status
T1	T_G explicitly defined: L_G, T_G, Models(T_G)	§2.4–§2.5: L^+, axioms (O$_1$)–(O$_3$), admissible class $\mathcal{K}$	✓
T2	Two enriched structures $M_1^G, M_2^G \models T_G$ exist	Proposition 2.4 construction; script `gate15_check.py`	✓
T3	Same L_0-reduct: $M_1^G \upharpoonright L_0 = M_2^G \upharpoonright L_0$	By construction in Proposition 2.4; Lemma 2.3; script `gate15_check.py`	✓
T4	Different G: $G^{M_1} \neq G^{M_2}$	By construction in Proposition 2.4; script `gate15_check.py`	✓
T5	Both structures satisfy the constraints Γ	Remark below (the constructed pair satisfies Γ in both members); script `gate15_check.py`	✓
T6	Result statable independently of $G \notin L_0$ (non-trivial)	Proposition 2.6 (stability under arbitrary L_0-theories)	✓
T7	Definability application formally valid	Definitions 2.7–2.8, Proposition 2.5, §2.6.1	✓
T8	E1/E2 comparison complete: representability vs. adequacy	§2.13 (minimal-enlargement benchmark)	✓
T9	Sensitivity test across L_0^A, L_0^B performed	§2.12; script `encoding_sensitivity_check.py`	✓
T10	Machine verification separated from proof	§2.10: the scripts are reproducibility artefacts; the general argument is Lemma 2.3	✓

Table 1: Gate 1.5 — non-triviality / recoverability check, 10/10.

The ten items are verified individually; the complete check is recorded in Table 1.

Items T1, T6, T7, T8 and T10 are discharged by proof or definition in the manuscript at the locations cited in the table. Items T2–T5 are verified computationally by `gate15_check.py` (frozen output `gate15_output.txt` in the reproducibility package), which reconstructs the Proposition 2.4 pair and asserts the admissibility axioms (O$_1$)–(O$_3$), the equality of the L_0-reducts, the difference of the G extensions, and the structural constraints Γ on both members. Item T9 is discharged by `encoding_sensitivity_check.py`: the outcome is encoding-sensitive in degree (L_0^A: undetermined for all 16 base assignments; L_0^B: undetermined for 6 of 10 admissible assignments) and robust in existence — under both encodings the extensional base alone fails to fix the modal content. The general thesis survives; the encoding-dependence of the strength of the residue is recorded as a finding (hermeneutic dependence), not concealed.

Two remarks complete the check. First, the model pair of Proposition 2.4 — G_1 containing $\langle \mathrm{custFact}(b), \mathrm{nonjustFact}(b, c)\rangle$ and G_2 empty — satisfies the standard structural constraints Γ (irreflexivity, asymmetry, transitivity) in both members: neither relation contains a reflexive pair, neither contains a pair and its converse, and neither contains a non-trivial chain. This is verified by `gate15_check.py` (item T5). The underdetermination therefore does not trade on a pathological choice of G, even though the admissible class $\mathcal{K}$ deliberately imposes no structural principles on G (condition 4 of §2.5). Second, T10 is a standing discipline of §2.10: the finite Boolean enumeration, the model-pair script, and the encoding-sensitivity enumeration are reproducibility artefacts, and the general arguments are supplied by Lemma 2.3 and the propositions cited in Table 1. The section does not proceed to historical inference until all ten items are discharged.

2.15 Hyperintensionality: The Four-Layer Claim

The formal section can be read as a single four-layer claim about hyperintensionality. The layers are labelled HI1–HI4 to avoid confusion with the Humean readings H1–H3 of §3.

HI1. *Literature.* Grounding and explanation are widely treated in the contemporary literature as hyperintensional phenomena (SEP "Metaphysical Grounding"; Fine 2012; Schnieder 2011).

HI2. *Formal specification.* The extensional language L_0 does not contain explanatory dependence as a primitive semantic resource (§2.2).

HI3. *Model-theoretic result.* Holding the L_0-reduct fixed, the explanatory enrichment can vary: two admissible structures with identical L_0-reducts realize different grounding readings (Proposition 2.4).

HI4. *Philosophical interpretation.* The extensional material profile does not by itself determine the explanatory structure of the target interpretation.

The chain runs Literature → Formalization → Theorem → Interpretation. The point of separating the layers is to block the leap from "grounding is hyperintensional" to "first-order logic cannot express grounding": HI1 is a claim about the literature, HI2 about a specified signature, HI3 a theorem about that signature, and HI4 an interpretation of the theorem. The present result occupies layers HI2–HI4; it neither needs nor asserts the unrestricted version of HI1 as a theorem about the expressive power of first-order logic. This is the formal counterpart of the distinction drawn in §2.13 between representability and adequate representation, and of the modal/grounding distinction $\Box(Custom(b) \to \neg Just(b, c)) \neq G(\text{custFact}(b), \text{nonjustFact}(b, c))$ of §2.7.

3 Hume and the Principle of Sufficient Reason: A Qualification Required

A widespread reading says "Hume rejects the metaphysical PSR"; another says "Hume does not reject the PSR; he merely re-describes the question". Neither is correct on its own. The verifiable facts are these.

(a) Hume denies that the causal maxim is rationally demonstrable. In Treatise 1.3.3, the maxim "whatever begins to exist must have a cause of existence" is said to be "neither intuitively nor demonstrably certain": its contrary is conceivable, and our assent to it arises from observation and experience, i.e., from custom (T 1.3.3; SBN 78–82). The denial is therefore epistemic: the maxim is not rationally grounded.

(b) Hume explicitly denies absolute or metaphysical necessity in causal connection. T 1.3.14.35: "there is no absolute nor metaphysical necessity, that every beginning of existence shou'd be attended with such an object" (SBN 172).

(c) Hume's positive account is psychological. T 1.3.14.20: necessity "is nothing but an internal impression of the mind, or a determination to carry our thoughts from one object to another" (SBN 164–65). The same thesis is stated in the Enquiry: "When we look about us towards external objects, and consider the operation of causes, we are never able, in a single instance, to discover any power or necessary connexion" (E 7.6; SBN 63). And the famous remark that "reason is nothing but a wonderful and unintelligible instinct in our souls" (T 1.3.16.9 SBN 178–9) makes the process-based character of the account explicit: belief formation is a mechanism, not an inference.

Hence: Hume does not "refute" the PSR in its causal form, and he does not leave it untouched; he removes it from the level of demonstration and relocates it at the level of custom. The reading "Hume denies the PSR" is false if it means he argued against the principle's content; the reading "Hume never questions it" is false if it means he accepted it as a rational principle. The correct formulation is: the causal maxim is, for Hume, a natural psychological regularity masquerading as a rational necessity.

3.1 The four-level Humean analysis

The formal treatment of §2 operates on several levels of claim at once. To keep the formal result from silently importing a contested interpretive thesis, we record explicitly the four-level decomposition on which it relies, tagging each level with its textual anchoring and its standing in the literature (see also §2.8 of the formal section).

Level	**Claim**	**Status**
Psychological-genetic	Causal expectations are formed by custom/habit, not by demonstrative reason.	Directly defensible from Hume's text (T 1.3.14.20, SBN 164–65; E 7.6, SBN 63; T 1.3.16.9, SBN 178–9).
Epistemic	These beliefs are not demonstratively justified.	Defensible, but the term "justification" must be unpacked (§2.2.2).
Normative exclusion	Custom-production excludes rational justification.	Contested contemporary reconstruction; alternative readings exist (H-II, naturalistic; Della Rocca 2010 on PSR as demotion vs. rejection; Pruss 2006 on Hume as evaporationist).
Grounding	The custom-production relation *grounds* the absence of justification.	Contemporary regimentation in the sense of Fine (2012) and Schnieder (2011); not directly attributable to Hume.

We do not attribute a contemporary grounding relation, nor a fully articulated theory of epistemic justification, to Hume. We use grounding only to regiment one strong contemporary reconstruction of the explanatory force that an interpreter may take Hume's appeal to custom to possess; alternative reconstructions remain live.

Two further points. First, the term "Principle of Sufficient Reason" (*principium rationis sufficientis*) is Leibnizian (*Monadologie* §32; 1714); Hume never uses it, and retro-application requires care. Second, the literature on Hume and the causal maxim is substantial: on the maxim itself, see Garrett 1997, 24, 66–69; on the projectivist reading of causal necessity, Beebee 2006; on the Enquiry context, Millican 2002, 27–65. The contemporary revival of PSR debates (Della Rocca 2010; Pruss 2006) has made the question of what exactly Hume denied once more a live interpretive issue; our claim (a)–(c) is deliberately minimal and textually anchored.

4 The Transmission of Stoic Epistemology to Early Modern Europe: A Verifiable Route

4.1 Sextus Empiricus

The kataleptic impression and the criterion debate are documented most fully in Sextus Empiricus, *Adversus Mathematicos* VII: the criterion discussion at M 7.24–29, the definitional texts at M 7.248 and 7.402, and the "clear, evident, striking" (*enarges*, *ektypos*, *plektike*) marks at M 7.257–58 (cf. Bury 1935, 7.402–8). The printing history is now well documented (Floridi 2002; Popkin 1979, 18): the first Latin translation of the *Hypotyposes* was published in 1562 by Henri Estienne (Geneva); the Latin *Adversus Mathematicos*, together with a second edition of the Estienne translation, was published in 1569 by Gentian Hervet (Paris: Martin Juvenis; Antwerp: Plantin). The Greek *editio princeps* of Sextus' *Opera* appeared only in 1621 (Geneva: Chouët). The common claim that "Hervet translated Sextus in 1562 and 1569" therefore conflates Estienne with Hervet and should be corrected.

4.2 Diogenes Laertius

The main doxographical source for Zeno's and Chrysippus' epistemology (the definitions of *phantasia*, *phantasia kataleptike*, *katalepsis*, *epistēmē* at *Vitae Philosophorum* 7.46–54; the criterion at 7.54 [Dorandi 2013]; cf. Bobzien 2003, 85–123) reached early modern Europe through Ambrogio Traversari's Latin translation, completed in manuscript by 1433; the Latin *editio princeps* was printed in Rome 1472 (Georgius Lauer); the celebrated Venice 1475 (Nicolaus Jenson) is the second printed edition; the Greek *editio princeps* appeared in Basel 1533 (Froben) (Dorandi 2013, 15; ISTC id00219000/id00220000).

4.3 Cicero and the Neostoic Reception

Cicero's *Academica* is the Latin source for Zeno's fist analogy — open hand: impression; fingers bent: assent; fist: katalepsis; the other hand grasping the fist: *epistēmē*, available to the sage alone (*Lucullus* 145 [Brittain 2006]; SVF 1.66) — and for the criterion debate in Latin. The *Academica* entered print early: *editio princeps* Rome 1471 (Sweynheym & Pannartz), with the Aldine edition of 1523 establishing the standard text (Hunt 1998). But the decisive early modern channel for Stoic doctrine as a system was Neostoicism: Justus Lipsius published *De Constantia* (Antwerp: Plantin, 1584), and in 1604, both the *Manuductio ad Stoicam philosophiam* and the *Physiologia Stoicorum* (Antwerp: Plantin-Moretus). The reception of Stoic ethics through Lipsius and du Vair is well documented (Lagrée 1994; du Vair 1594); what matters for this note is that the epistemological transmission — the katalepsis doctrine — had to proceed through Sextus, Diogenes Laertius and Cicero, since the Neostoics' interest was primarily ethical.

4.4 Hume's Own Access

For Hume's own acquaintance with these materials, the documented facts are these. The catalogue of Hume's library (Norton & Norton 1996) lists Cicero's *Opera* (Paris, 1740–42, 9 vols.); it does not list Sextus Empiricus. The 1840 sale catalogue, on which the reconstruction rests, shows no Sextus (Fosl 1998). However, the 1840 catalogue reflects books sold at auction; it is not identical to the full library at Hume's death, as books were often given away or sold earlier. The absence of Sextus is therefore negative evidence, not a proof of Hume's ignorance. But building a "Hume owned a Sextus" narrative on this gap is historiographically irresponsible. The standard scholarly account remains: Hume's access to Pyrrhonism ran through Bayle, Montaigne and others (Popkin 1952, 65–81; 1951). The historiography of "Hume the Pyrrhonian" is best grounded in: (i) Hume's direct reading of Cicero; (ii) his mediated reading of Sextus through the French sceptical tradition.

4.5 The Medieval Question and the Cartesian Comparison

Linear chains of the form "Stoicism → Aristotelian commentators → Scholasticism → Descartes → Locke" are not documented. The evidence for discontinuity is strong: only seven Latin Sextus manuscripts circulated in the medieval Latin West (Floridi 2002); the *Academica* was among the rarer Cicero texts in the Middle Ages, with Henry of Ghent the first scholastic explicitly to engage it (Schmitt 1972; 1983; SEP "Medieval Skepticism"); the term "scepticus" entered Latin usage only in the 1430s. Two qualifications are required, in the spirit of §8. First, "no continuous direct transmission of the doctrine of katalepsis" does not mean "no knowledge of Stoic epistemology at all": Cicero's *comprehensio* and Augustine's *Contra Academicos* provided an indirect, contested channel (Nawar 2022, 185–207). Second, the relation between Descartes's "clear and distinct ideas" and the kataleptic impression should be studied as discrete responses to a common skeptical problem — the criterion problem — rather than as a transmission chain; this is the position of the relevant secondary literature (Frede 1983; Nawar 2022).

Similarly, Locke's critique of innate principles in Essay I.3 ("No Innate Practical Principles") is directed, in its explicit form, at Herbert of Cherbury's *De Veritate* (1624): Herbert is named at I.3.15 and discussed through I.3.27 (Nidditch ed. 1975, 77–84); Descartes is not the primary target of that chapter. And the very notion of a unified "British Empiricism" is a construction of nineteenth-century historiography (Norton 1981). The moral of this section: the history of philosophy should be written from the history of printed texts, not from speculative chains.

4.6 Historical-Evidential Method: The Ev0–Ev4 Ladder

The historical discipline of §4 can be made explicit as an evidence ladder. The labels are Ev0–Ev4 to avoid confusion with the expressive-enrichment levels E0/E1/E2 of §2.13.

Ev0 *Bibliographic presence*: the text exists and is attested.

Ev1 *Documented access*: an agent or library can be shown to have had the text available.

Ev2 *Documented citation*: the text is cited or quoted by the agent.

Ev3 *Textual dependence*: the agent's claims require the text's specific content.

Ev4 *Influence*: the text shaped the agent's doctrine.

Rule: $\mathrm{Ev}_n \not\Rightarrow \mathrm{Ev}_{n+1}$ in general. Two prohibitions follow, which the present note observes throughout. First, *availability is not influence*: that Sextus was printed in 1562 does not show that Hume's doctrine was shaped by him. Second, *citation is not dependence*: that an author cites a text does not show that the cited content is load-bearing for the author's argument. The transmission route of §4.1–3 is established at the level of Ev1–Ev2 (documented access and citation); Hume's own acquaintance with Sextus is at most Ev1–Ev2 via Bayle and the French sceptical tradition (§4.4); and the medieval chain is rejected because it does not reach Ev1 (§4.5). No claim of Ev4 is made anywhere in this note.

5 A Comparative Note: Formal Limits Across Philosophical Traditions

The theme of this note — that a formal scheme has limits that must be stated from within and beyond — is not peculiar to the Western analytic tradition. We record three documented parallels and note two further items for completeness. Each is invoked as an analogy only: no claim of historical transmission between these traditions and the Stoic/Humean material is made or implied (cf. §8).

(a) **The catuşkoṭi.** The Buddhist four-cornered negation schema — P, $\neg P$, both, neither — was theorized by Nāgārjuna as an exhaustive decomposition of assertoric space, and it has been shown that classical two-valued semantics cannot express the fourth corner as a distinct position; the schema is naturally formalized in paraconsistent logics (Priest 2010, 24–54; 2018). One asymmetry must be stated honestly: the catuşkoṭi is a case of expanding a formal scheme — paraconsistency extends the logic — whereas our §2 result is a case of a scheme's representational limit. The parallel is therefore not in the kind of limit but in the practice: the catuşkoṭi shows that theorizing about what a scheme can and cannot say — including the scheme's own boundary — is a distinguished non-Western philosophical activity. Whether Nāgārjuna himself intended the catuşkoṭi as a meta-logical reflection on the limits of assertoric space, or whether this is a modern reconstruction (Priest 2018, ch. 1; Tillemans 1999), is itself a contested interpretive question. The claim is not that Stoic or Humean materials require paraconsistency; whatever Nāgārjuna's intentions, the catuşkoṭi schema itself — as an exhaustive decomposition of assertoric space — encodes a boundary-theorizing practice, and so the question "what cannot be represented here?" is not a modern Western invention.

(b) **The Xunzian rectification of names (zhengming).** Xunzi's chapter "On the Rectification of Names" (Xunzi 22) addresses the criterion problem for language: names are correct if fixed by shared sensory experience (the *tianguan*, "heavenly faculties", common to all humans — Xunzi 22.5) and by convention ("names have no intrinsic appropriateness; they are fixed by agreement" — Xunzi 22.9; Hansen 1992, ch. 9; Hansen 1983). The structural parallel with the Stoic criterion debate is striking and has been noted in the comparative literature (Rošker, SEP "Epistemology in Chinese Philosophy"; Kjellberg 1996): both theories face the question of what fixes the standard by which representations are judged correct. The differences are equally instructive. At the level of the kind of standard, the Stoic criterion is source-based (the impression's provenance) while the Xunzian criterion is intersubjective-conventional. But the deeper difference is in the functional position each doctrine occupies within its system. The Stoic katalepsis is an epistemological doctrine about individual cognitive accuracy: it serves to secure truth for a single cognitive agent, and its failure mode is epistemic error. The Xunzian zhengming, by contrast, is an ethical-political doctrine about the ordering of social life: its purpose is to ensure that correct naming yields correct action and social coordination (cf. Xunzi 22.9; the canonical formula "if names are not correct, language will not be in accordance with truth" is *Analects* 13.3 — the Xunzian claim is a development in the same spirit rather than a verbatim citation), and its failure mode is social disorder. The structural parallel at the level of “what fixes the standard?” thus masks a difference in the domain and the end to which the standard is put: the analogy should not be read as implying that the two doctrines occupy the same functional position in their respective systems. A first-order encoding of either faces the same underdetermination problem we isolate in §2.6: the criterion’s normative force — in virtue of what the standard is the standard — is not a truth-condition.

(c) **Kaozheng and documentary discipline.** The Qing-dynasty kaozheng (*kaoju*; “evidential research”) movement — exemplified by Dai Zhen (*Mengzi ziyi shuzheng*, Preface; in *Dai Zhen wenji*, Zhonghua Shuju, 1980; cf. Elman 1984, Introduction and ch. 5; 2nd ed. 2001) — made textual verification, philological evidence and the rejection of unfounded commentary the methodological core of scholarship. Our §4 and §8 defend precisely this discipline for the history of philosophy: the claim that documentary discipline is a method, not an ornament, has a systematic analogue in a non-Western tradition that is still under-cited in analytic historiography. The parallel is in the methodological attitude — evidence over speculation — not in the object of inquiry, which differs. Again: analogy, not influence.

We also note, for completeness, the Mohist tradition’s treatment of the name–reality (*ming*/*shi*) distinction and the argumentation chapter “Lesser Selection” (*Xiaoqu*) of the *Mozi* (Graham 1978; 1989), and the White Horse paradox of Gongsun Long, which turns on the gap between a predicate’s linguistic ontology and its extension — a theme in the vicinity of, though not identical with, the model-theoretic underdetermination discussed in §2.6 (Hansen 1983, ch. 5). Finally, Wang Yangming’s doctrine of *liangzhi* (“innate knowing”; *Chuanxilu* 2.2; Van Norden, SEP “Wang Yangming”) treats the cognitive standard as an inner, quasi-evidential faculty rather than an external impression or convention — structurally closer to the Stoic katalepsis than to the Xunzian convention, since both posit a source-based standard rather than an intersubjective one, yet differing in that Wang Yangming’s doctrine is embedded in a moral-spiritual project (the unity of knowledge and action) rather than primarily in an epistemological criterion debate. [Provenance note: the chapter and passage numbers above follow the standard editions cited in the bibliography; where translations differ (Watson; Knoblock), we rely on the philological consensus reported in Hansen 1992 and the Stanford Encyclopedia entries cited. The claim that the Stoic/Xunzian parallel has been noted in the comparative literature is supported by Kjellberg 1996 (Sextus/Zhuangzi/Xunzi on skepticism) and by the SEP entry “Epistemology in Chinese Philosophy” (Rošker); both are cited as reporting a parallel, not as establishing a historical connection.]

6 Objections and Replies

The central result invites a family of objections. They are collected here following the format *objection – concession – distinction – response.* No reply introduces new machinery; each restates the scope fixed in §2.11 and §2.14.

Objection 1 (“G is merely a new predicate”). Adding a primitive G is something any first-order language can do; the result therefore shows only that a symbol absent from L_0 is absent from L_0.
Concession. Yes: first-order logic can accommodate a primitive binary relation G.
Distinction. That G can be *added* does not show that it is *definable from* L_0, nor that the resulting predicate has the semantic properties characteristic of grounding.
Response. The claim concerns recoverability from a specified extensional base, not the impossibility

of adding vocabulary; the non-triviality check of §2.14 (item T6) is exactly the point on which the objection turns.

Objection 2 ("Modal logic reduces to FOL"). The standard translation embeds modal logic in first-order logic; the modal residue is therefore not a genuine residue.
Concession. The standard translation is a genuine embedding for the model-theoretic content of modal logic.
Distinction. The standard translation preserves *truth in a model*; it does not preserve explanatory direction. In particular, $\Box(Custom(b) \to \neg Just(b,c))$ ("it is necessary that ...") is not the grounding claim $G(\text{custFact}(b), \text{nonjustFact}(b,c))$ ("... because ...").
Response. The modal enrichment captures necessity, but not by itself the explanatory direction; §2.7.1 states this explicitly.

Objection 3 ("Adding G to FOL solves the problem"). If G can be added, the problem is solved; the paper's own E1 shows this.
Concession. E1 makes the relation nameable.
Distinction. Naming is not yet adequacy: a primitive G is not a grounding theory. The constraints Γ of E2 are required for the relation to be characterized as grounding, and Γ is itself an interpretive choice (§2.13).
Response. The minimal-enlargement benchmark (E0/E1/E2) is precisely the experiment that separates "added vocabulary" from "added theory".

Objection 4 ("A primitive grounding relation suffices"). Grounding may simply be taken as primitive; no theory is needed.
Concession. A primitive relation can represent a formal relation.
Distinction. That the primitive relation *is* grounding — and not a merely similar relation — is a further semantic claim.
Response. The present result does not deny the possibility of a primitive G; it denies that G is determined by the L_0-reduct, and it records that characterizing G as grounding requires the semantic constraints of E2.

Objection 5 ("The result is a semantic triviality"). The result states only that a relation absent from the signature is not determined by that signature.
Distinction. Trivial lexical absence ($G \notin L_0$) is not semantic underdetermination ($\mathcal{M}_1 \upharpoonright L_0 = \mathcal{M}_2 \upharpoonright L_0$ while $G^{\mathcal{M}_1} \neq G^{\mathcal{M}_2}$). The latter is what Proposition 2.4 establishes.
Response. The Gate 1.5 check (§2.14, item T6) is designed to make this distinction enforceable rather than asserted.

Objection 6 ("The Stoic reconstruction is underdetermined"). The Stoic material is encoded one way; a different encoding would give a different result, so the conclusion is an artefact of the choice.
Concession. The choice of encoding is interpretive and precedes the formal argument (§8).
Distinction. The relevant question is whether the negative result survives across more than one reasonable encoding.
Response. The encoding-sensitivity test of §2.12 (L_0^A vs. L_0^B) is the honest procedure for answering this; its outcome (robust or sensitive) is reported rather than assumed (Gate 1.5, item T9).

Objection 7 ("The Hume reconstruction is controversial"). The demotion reading (H2) is not the only interpretation of Hume's appeal to custom; the result therefore inherits a contested premise.
Concession. H1 (strong denial) and H3 (evaporation) are live alternatives in the literature.

Distinction. The formal result is invariant under the choice between the exclusion readings modulo the annotations of §2.2.2; the four-level analysis of §3.1 is tagged by textual anchoring and standing. *Response.* H2 is adopted as a methodological interpretation for reconstruction, not as a settled historical conclusion (§3).

6.1 What the Result Does Not Show: A Negative-Result Matrix

The replies above can be summarized in a matrix that pairs each claim with what is *not* shown and what *is* shown. The matrix doubles as an editorial guard: any sentence that asserts a "not shown" claim as a result is a scope violation.

Claim	**What is not shown**	**What is shown**
FOL fails	Intrinsic impossibility	L_0-relative underdetermination
Grounding impossible in FOL	—	Primitive extension possible
G represents grounding	Not by symbol alone	Requires semantic/theoretical constraints Γ
Modal logic solves the explanatory problem	—	Captures necessity, not explanatory direction
Historical reading follows from the formal model	—	Requires independent documentary evidence (§4)
Richer language cannot represent the distinction	—	Representation requires an additional semantic resource

7 Conclusion

This note has made three contributions.

1. **Technical limit.** A purely extensional first-order encoding of the justification conditions of Stoic katalepsis and Humean custom can represent the material content of both positions — including the extensionally distinguishable formulations T_1 and T_2 (formerly (Exc1) and (Exc2)) — but cannot fix their explanatory and modal residue: the "because" of the Humean exclusion (hyperintensional; grounding), the "could not arise from what is not" of the Stoic definition (modal), and the reliability of belief-forming processes. More precisely, the grounding atom $G(\mathrm{custFact}(b), \mathrm{nonjustFact}(b, c))$ is *not implicitly definable* in the extensional signature L_0 over the admissible class of reconstructions $\mathcal{K}$ (Proposition 2.4); the model-pair argument shows the limit is semantic: the intended reading is not a structural property of L_0-models over $\mathcal{K}$. A justification operator (Artemov & Fitting 2019), a grounding connective (Fine 2012), or a modal operator is required. This is a singular encoding limit, not a universal failure claim.

2. **Interpretive limit.** Hume's relation to the PSR is neither simple rejection nor silence: he relocates the causal maxim from the demonstrative level to the level of custom (T 1.3.3; SBN 78–82; T 1.3.14.20; SBN 164–65; T 1.3.14.35; SBN 172). "Hume never questions the PSR"

and "Hume refutes the PSR" are both false; the accurate formulation is a demotion, not a refutation. A significant interpretive counter-thesis — defended notably by Della Rocca (2010) and, in a different register, by Pruss (2006) — argues that this "demotion" is a polite fiction: by reducing the PSR to a "natural psychological regularity," Hume effectively evaporates its normative status, rendering the distinction between "demotion" and "rejection" functionally moot. We maintain the demotion formulation as the more textually anchored reading, since Hume explicitly preserves the causal maxim as a psychological principle rather than arguing against its content; but the evaporation reading remains a strong interpretive alternative, and our claim should be read as a preference supported by the textual evidence, not as a settled result.

3. **Historical limit.** The transmission of Stoic epistemology to early modern Europe runs through documented printed texts: Sextus (Latin 1562 Estienne; 1569 Hervet; Greek 1621), Diogenes Laertius (Traversari 1433; Latin *editio princeps* Rome 1472; Venice 1475; Greek Basel 1533), Cicero's *Academica* (in *Opera omnia*, Rome 1471 [Sweynheym & Pannartz]; Aldine edition 1523; the fist analogy at *Lucullus* 145 [Brittain 2006]), and the Neostoic reception (Lipsius 1604). Speculative medieval chains are not documented; indirect medieval knowledge via Cicero and Augustine is. Hume's own access ran through Cicero directly and Sextus through Bayle (Popkin 1952, 65–81).

Formal tools are useful precisely when their limits are stated; the history of philosophy requires a documentary discipline that formal notation cannot supply; and both theses have systematic, independently developed analogues in the Buddhist, Xunzian and kaozheng traditions (§5).

8 Limits of This Note

In the same spirit, the limits of this note are stated explicitly.

1. We make no claim of in-principle inexpressibility for any language richer than L_0: the negative result is about this signature, at this granularity, under extensional semantics, over the admissible class $\mathcal{K}$.

2. We do not claim that justification logic, grounding, or modal logic "solves" the philosophical problems; we claim only that they make the relevant distinction representable.

3. The comparative section (§5) is analogical; none of its claims depends on a historical connection between the traditions compared, and none asserts one.

4. All historical claims rest on the printed-text history summarized in §4; where the secondary literature is the sole support (e.g., the medieval manuscript census), this is stated, and the sources are given.

5. "No evidence of continuous direct transmission" (§4.5) is a negative-evidence claim; it is recorded as such and is not asserted to be a proof of non-transmission.

6. Quotations follow the editions listed under "Citations and Editions"; paragraph numbers for Hume follow the Norton & Norton (2000) and Beauchamp (1999) editions, with Selby-Bigge/Nidditch (SBN) pagination given in every case.

7. The formal encoding in §2.2 is itself an interpretive choice. Different readings of the Stoic and Humean sources might yield different signatures or axioms; our result is relative to the encoding chosen, and the choice of encoding is a hermeneutic decision that precedes the formal argument.

8. The richer frameworks of §2.7 — grounding, justification logic, process reliabilism — are invoked as contemporary regimentations of the relevant claims; no claim is made that Hume or the Stoics held these theories, and their vocabulary is applied in a deliberately anachronistic, analytic sense.

9 Appendix: Definitions

Term	Definition	Source
katalepsis	Stoic "cognition": the grasping of a kataleptic impression; the criterion of truth for the Stoics	DL 7.46, 7.54 [Dorandi 2013]; translation "cognition": Long & Sedley 1987, 40; SEP "Stoicism" §3.7
phantasia kataleptike	Kataleptic impression: "arising from what is, stamped and impressed in accordance with what is, of such a kind as could not arise from what is not"	DL 7.46 [Dorandi 2013]; M 7.248 [Bury 1935]; 7.402 [Bury 1935]
Criterion problem	The question what standard fixes which impressions are true; the regress of "criterion of the criterion"	Sextus M 7.24–29; PH I.164–169 (169 = diallelos); M 7.440–446
Assent (sunkatathesis)	The Stoic act of approving an impression; the middle stage between impression and cognition	DL 7.46–54; Cicero, *Acad.* 2.37–39, 2.66–67
katalepsis / episteme	The Stoic hierarchy: katalepsis is the assent to a kataleptic impression (sunkatathesis kataleptikei phantasiai); episteme is secure cognition (katalepsis asphales)	M 7.151–152 [Bury 1935]; DL 7.47 [Dorandi 2013]
PSR	*Principium rationis sufficientis*: Leibnizian term; every fact has a sufficient reason	Leibniz, *Monadologie* §32 (1714)
FOL	First-order logic: extensional, truth-functional model-theoretic semantics	Standard; §2.2
L_0	The minimal many-sorted extensional language of the formal section: sorts I (impressions), $Cont$ (contents), B (cognitive episodes); predicates Kat, Rep, Grasp, Assent, Bel, Causal, Custom, Just, StoicEp	§2.2
L^+	The enriched language extending L_0 with reified fact-objects $Fact \subseteq Cont$, the functions $\mathrm{custFact} : B \to Fact$ and $\mathrm{nonjustFact} : B \times Cont \to Fact$, the grounding relation $G \subseteq Fact \times Fact$, and the obtaining monad Obtains	§2.4

$\mathcal{K}$	The class of admissible reconstructions: L_0-structures satisfying M_0 and the sortal and reification axioms O_1–O_3, with no bridge axiom and no structural principles on G	§2.5
Grounding	Hyperintensional explanatory relation: $\phi < \psi$, "ϕ grounds ψ"; in this paper regimented on reified fact-objects ($G \subseteq \mathit{Fact} \times \mathit{Fact}$)	Fine 2012; Correia & Schnieder 2012; §2.4
Implicit definability	A relation R is implicitly definable in L over a class $\mathcal{K}$ iff any two admissible structures sharing an L-reduct agree on R	Beth 1953; §2.6.1
Custom	Humean habit: determination of the mind to pass from one object to its usual attendant	T 1.3.14.20; E 5
Minimal dependency skeleton	The Stoic reconstruction (S): $\forall b\,(\mathrm{StoicEp}(b) \to \exists i \exists c\,[\mathrm{Kat}(i) \wedge \mathrm{Rep}(i,c) \wedge \mathrm{Grasp}(b,i) \wedge \mathrm{Assent}(b,c)])$; deliberately excludes truth, guarantee and the modal clause	§2.2.1
Four Humean levels	Psychological-genetic / epistemic / normative exclusion / grounding; see §3.1	§3.1
catuşkoṭi	Buddhist four-cornered negation: P, $\neg P$, both, neither	Priest 2010
zhengming (*zheng ming*)	Xunzian rectification of names: fixing correct usage by sensory agreement and convention	Xunzi 22.5, 22.9; Hansen 1992
kaozheng (*kaoju*)	Qing evidential scholarship: philological verification as method	Elman 1984, Introduction and ch. 5

Citations and Editions

- Hume, *Treatise* (T): Norton & Norton (eds.), Oxford University Press, 2000; paragraphs cited as T 1.3.3 etc.; Selby-Bigge/Nidditch (SBN) pagination in parentheses.
- Hume, *Enquiry* (E): Beauchamp (ed.), Oxford University Press, 1999; paragraphs cited as E 7.6 etc.; SBN pagination in parentheses.
- Sextus Empiricus: Primary source editions: Latin translations — *Hypotyposes*, tr. Henri Estienne, 1562; *Adversus Mathematicos*, tr. Gentian Hervet, 1569; Greek *Opera*, Chouët, 1621. Modern translation used: Bury (tr.), Loeb Classical Library, 1933–49.
- Diogenes Laertius: Hicks (tr.), Loeb Classical Library, 1925; Dorandi (ed.), Cambridge University Press, 2013.
- Cicero, *Academica*: Brittain (tr.), Hackett, 2006; *Lucullus* paragraph numbers (Plasberg).
- Dai Zhen: *Mengzi ziyi shuzheng* (*Mengzi ziyi shuzheng*), in *Dai Zhen wenji* (*Dai Zhen wenji*), Beijing: Zhonghua Shuju, 1980.
- von Arnim, H. (ed.). *Stoicorum Veterum Fragmenta* (SVF), vol. 1. Leipzig: Teubner, 1905.

Provenance Table

§	Claim	Source
Abstract	katalepsis = "cognition" (Long & Sedley 1987, 40)	Long & Sedley 1987, 40; cf. SEP "Stoicism" §3.7 [P-01]
§2.9	Kataleptic definition with modal clause	DL 7.46 [Dorandi 2013]; M 7.248 [Bury 1935]; 7.402 [Bury 1935] [P-03]
§9	katalepsis = sunkatathesis kataleptikei phantasiai; episteme = katalepsis asphales	M 7.151–152 [Bury 1935]; DL 7.47 [Dorandi 2013] [P-03b]
§2.3	$T_2 \wedge M_0 \models T_1$; $T_1 \wedge M_0 \not\models T_2$; unique countermodel $(0, 1, 1, 1)$	Formal argument; Table A (16 Boolean rows) [P-04]
§2.3	Bridge collapse: $T_1 \wedge M_0 \wedge B_0 \models T_2$ (0 falsifiers)	Formal argument; machine-checked [P-04b]
§2.4	Grounding hyperintensional; regimented on reified fact-objects	Fine 2012; Correia & Schnieder 2012; Schnieder 2011 [P-05]
§2.6	Grounding atom not implicitly definable in L_0 over $\mathcal{K}$	Formal argument; Lemma (reduct invariance); Beth 1953 [P-05b]
§2.6.1	Explicit vs. implicit definability; Beth anchor	Beth 1953 [P-05c]
§2.7	Process reliabilism	Goldman 1979, 1–23; 1986 [P-05]
§2.7	Justification logic	Artemov 2008; Artemov & Fitting 2019 [P-06]
§3	T 1.3.3; SBN 78–82	Hume, T (Norton & Norton 2000); SBN [P-07]
§3	T 1.3.14.20; SBN 164–65	Hume, T [P-07]
§3	T 1.3.14.35; SBN 172	Hume, T [P-07]
§3	E 7.6; SBN 63	Hume, E (Beauchamp 1999) [P-07]
§3	T 1.3.16.9; SBN 178–9	Hume, T [P-07]
§3	PSR literature	Garrett 1997; Beebee 2006; Millican 2002; Della Rocca 2010; Pruss 2006 [P-07]
§4.1	1562 Estienne; 1569 Hervet; Greek 1621	Floridi 2002; Popkin 1979, 18 [P-08]
§4.1	M 7.24–29; M 7.257–58	Sextus, M (Bury, Loeb) [P-08]
§4.2	Traversari 1433; Rome 1472; Venice 1475; Basel 1533	Dorandi 2013, 15; ISTC id00219000/id00220000 [P-09]
§4.3	Fist analogy at *Lucullus* 145 [Brittain 2006]	Cicero, *Acad.* (Brittain 2006); SVF 1.66 [P-10]
Abstract	Fist analogy = Long & Sedley 41B	Long & Sedley 1987, section 41B (= Cicero, *Lucullus* 145) [P-01]
§4.3	Lipsius 1584; 1604	SEP "Justus Lipsius"; Lagrée 1994 [P-10]

§4.4	Hume's library: Cicero yes, Sextus no	Norton & Norton 1996; Fosl 1998 [P-11]
§4.4	Bayle-mediated access	Popkin 1952, 65–81; 1951 [P-11]
§4.5	Medieval discontinuity; nuances	Floridi 2002; Schmitt 1972; SEP "Medieval Skepticism"; Nawar 2022 [P-12]
§4.5	Locke–Herbert I.3.15–27	Locke, *Essay* (Nidditch 1975), 77–84 [P-12]
§4.5	British Empiricism myth	Norton 1981 [P-12]
§5	Catuşkoṭi formalized	Priest 2010; 2018 [P-13]
§5	Xunzi 22.5; 22.9	Xunzi (ctext; Knoblock tr.); Hansen 1992, ch. 9 [P-13]
§5	Kaozheng	Elman 1984, Introduction and ch. 5; 2nd ed. 2001 [P-13]
§5	Mohist/Gongsun Long	Graham 1978; 1989; Hansen 1983, 140–51 [P-13]
§8	Limits stated	This note [P-15]

SC-ID mapping (CHK-02). Every mandatory SC-ID corresponds to at least one section covered by the table: SC-001–004 → Abstract/Özet/Keywords; SC-005 → §1; SC-006–012 → §2.2–§2.7; SC-013–014 → §3; SC-015–017 → §4; SC-018 → §5; SC-019 → §7; SC-020–022 → Provenance/Citations/References; SC-023 → Appendix: Definitions; SC-024–030 → §4.4, §4.5, Appendix, §8 (optional items are covered where adopted).

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The initial draft was generated in 2026 with the assistance of an AI language model used for formalization and bibliography retrieval. All models, ISTC numbers, SBN citations, and Greek transcriptions were manually verified against primary sources and critical editions. The reproducibility package accompanying this manuscript (the formal model-check script, its frozen output, and the manifest) records the verification artefacts separately from the mathematical argument. The human author is responsible for the final content.

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Review Compilation

Stoic_Hume_Formal_Section_2026-08-17

Part I — Revised manuscript: ingiliz_empirizmi_v3.pdf (33 pages)
Part II — Original manuscript: original_manuscript.pdf (19 pages)

Separator page — not part of either manuscript.

The Limits of Formal Methods: Notes on the Formal Representation of Stoic Katalepsis and Hume’s Critique of Causation

Biçimsel Araçların Sınırı: Stoacı Katalepsis ile Humecu Nedensellik Eleştirisinin Biçimsel Temsili Üzerine Notlar

2026

Abstract

This paper is a methodological note, not a historical thesis. It examines a narrow question: what can and cannot be represented when Stoic *katalepsis* —the "cognition" (Long & Sedley 1987, 40) of a kataleptic impression —and Hume's account of causation are encoded in first-order logic (FOL)? We formalize the two positions at the level of their justification conditions and argue that a purely extensional FOL encoding can represent the *material content* of both positions but cannot represent their *explanatory* and *modal* residue: (i) Hume's claim that a causal belief is unjustified *because* it is produced by custom has the form of an explanatory (grounding) claim, and grounding is hyperintensional —it is not a truth-condition and therefore not part of FOL model theory; (ii) the ancient definition of the kataleptic impression contains a literal modal clause —the impression is "of such a kind as could not arise from what is not" (Diogenes Laertius 7.46; Sextus Empiricus, *Adversus Mathematicos* 7.248, 7.402) —which any FOL encoding must de-modalize. The two residues are not symmetric: the Stoic residue is modal and is recovered by adding a modal operator, while the Humean residue is explanatory and requires a hyperintensional connective (grounding); the negative result is that neither is a truth-condition of the extensional encoding. We show this by a model-pair argument: two epistemologies with identical FOL structures realize different explanatory readings, and no first-order theory in the unexpanded signature can separate them. We do not claim that formalization "necessarily" fails; we claim only that this encoding, in the unexpanded signature L, at this level of granularity, is semantically —not syntactically — underdetermined with respect to the intended normative reading, and that representing the distinction requires a justification operator (justification logic: Artemov & Fitting 2019), a grounding connective (Fine 2012), or a modal operator beyond FOL. We then correct two historiographical errors that frequently accompany such formalizations and add documented channels: (i) the claim that Hume rejects the Principle of Sufficient Reason must be qualified —Hume denies that the causal maxim is rationally demonstrable (T 1.3.3; SBN 78–82) and denies any "absolute or metaphysical necessity" in causal connection (T 1.3.14.35; SBN 172); (ii) the transmission of Stoic epistemology to early modern Europe ran through documented printed texts —Sextus Empiricus (Latin *Hypotyposes*, tr. Henri Estienne, 1562; Latin *Adversus Mathematicos*, tr. Gentian Hervet, 1569; Greek *Opera*, 1621), Diogenes Laertius (Traversari's translation completed 1433; Latin editio princeps Rome 1472; Venice 1475; Greek editio princeps Basel 1533), and Cicero's *Academica* (editio princeps Rome 1471; the fist analogy at *Lucullus* 145 = Long & Sedley 41B), supplemented by the Neostoic reception (Lipsius 1604) —not through a speculative medieval chain. The paper concludes that formal tools are useful precisely when their limits are stated, that the history of philosophy requires documentary discipline that formal notation cannot supply, and that this stance has systematic analogues in non-Western traditions of textual and logical rigor (the Buddhist *catuṣkoṭi*; the Xunzian theory of the rectification of names; the Qing *kaozheng* tradition), which we invoke as analogies, not as transmission claims.

Keywords: Stoicism; *katalepsis*; *phantasia kataleptike*; Hume; causation; Principle of Sufficient Reason; first-order logic; justification logic; grounding; transmission of ancient philosophy; catuṣkoṭi; zhengming; kaozheng

Özet

Bu makale tarihsel bir tez değil, **metodolojik bir nottur**. Dar bir soruyu inceliyoruz: Stoacı *katalepsis* (kavrayıcı izlenimin "kavranması"; Long & Sedley 1987'nin çevirisiyle "cognition") ile Hume'un nedensellik açıklaması birinci derece mantıkta (FOL) kodlandığında ne temsil edilebilir, ne temsil edilemez? İki konumu gerekçelendirme koşulları düzeyinde biçimselleştiriyor ve tamamen ekstansiyonel bir FOL kodlamasının her iki konumun **maddi içeriğini** temsil edebildiğini, ancak **açıklayıcı ve modal kalıntıyı** temsil edemediğini gösteriyoruz: (i) Hume'un "nedensel inanç, alışkanlık tarafından üretildiği *için* gerekçelendirilmemiştir" iddiası açıklayıcı (grounding) biçimindedir ve grounding hiperintansiyoneldir —truth-koşulu değildir, dolayısıyla FOL model teorisinin parçası değildir; (ii) kavrayıcı izlenimin antik tanımı lafzî (sözcüğü sözcüğüne, gerçek anlamda) bir modal kloz içerir —izlenim "olmayandan doğamayacak türden"dir (Diogenes Laertius 7.46; Sextus Empiricus, *Adversus Mathematicos* 7.248, 7.402) —ve her FOL kodlaması bu klozu de-modalize etmek zorundadır. Bunu bir model-çifti argümanıyla gösteriyoruz: özdeş FOL yapılarına sahip iki epistemoloji farklı açıklayıcı okumalar gerçekleştirir ve genişletilmemiş imzada ('L' imzasında) hiçbir birinci derece teori bunları ayırt edemez. Formalizasyonun "zorunlu olarak" başarısız olduğunu iddia etmiyoruz; yalnızca bu kodlamanın, bu ayrıntı düzeyinde, amaçlanan normatif okumaya göre **semantik olarak** (sözdizimsel değil) eksik belirlenmiş olduğunu ve ayrımın temsili için FOL ötesinde bir gerekçe operatörü (gerekçe mantığı: Artemov & Fitting 2019), bir grounding bağlacı (Fine 2012) veya bir modal operatör gerektiğini söylüyoruz. Ardından bu tür biçimselleştirmelere sıklıkla eşlik eden iki tarihsel hatayı düzeltiyor ve belgelenmiş kanalları ekliyoruz: (i) Hume'un Yeter Sebep İlkesi'ni reddettiği iddiası nitelendirilmelidir —Hume, nedensel özdeyişin rasyonel olarak gösterilemeyeceğini (T 1.3.3; SBN 78–82) ve nedensel bağlantıda hiçbir "mutlak ya da metafizik zorunluluk" olmadığını (T 1.3.14.35; SBN 172) söyler; (ii) Stoacı epistemolojinin erken modern Avrupa'ya aktarımı belgelenebilir basılı metinler üzerinden gerçekleşmiştir —Sextus Empiricus (Latince *Hypotyposes*, çev. Henri Estienne, 1562; Latince *Adversus Mathematicos*, çev. Gentian Hervet, 1569; Yunanca *Opera*, 1621), Diogenes Laertius (Traversari'nin çevirisi 1433'te tamamlandı; Latince editio princeps Roma 1472; Venedik 1475; Yunanca editio princeps Basel 1533) ve Cicero'nun *Academica*'sı (editio princeps Roma 1471; yumruk benzetmesi *Lucullus* 145 = Long & Sedley 41B) ile Neostoacı alımlanış (Lipsius 1604) —spekülatif bir ortaçağ zinciri üzerinden değil. Makale, biçimsel araçların sınırları belirtildiğinde yararlı olduğu, felsefe tarihinin biçimsel notasyonun sağlayamadığı bir belge disiplini gerektirdiği ve bu tutumun Batı dışı geleneklerde de sistematik karşılıklarının bulunduğu (Budist *catuṣkoṭi*; Xunzici adların düzeltilmesi teorisi; Qing *kaozheng* geleneği) sonucuna varır —bu karşılıklar aktarım iddiası değil, analoji olarak anılır.

Anahtar Kelimeler: Stoacılık; *katalepsis*; *phantasia kataleptike*; Hume; nedensellik; Yeter Sebep İlkesi; birinci derece mantık; gerekçe mantığı; grounding; antik felsefenin aktarımı; catuṣkoṭi; zhengming; kaozheng

1 Introduction: Scope and Limits

The purpose of this note can be summarized in three points.

1. To encode the justification conditions of Stoic *katalepsis* and Hume's account of causation at first-order level, and to state precisely which distinction this encoding misses —not the material content of the two positions, but their explanatory and modal residue (§2).
2. To correct two historiographical errors that frequently accompany such formalizations —the unqualified claim about Hume and the Principle of Sufficient Reason (§3), and the speculative reconstruction of the transmission of Stoic epistemology (§4) —and to add the documented channels and the relevant secondary literature.
3. To avoid universal claims such as "formalization necessarily fails" and to state only singular, verifiable limits —of this encoding, at this level of granularity —together with a statement of what a richer logical language would add (§2.4).

A brief comparative note (§5) places the theme of the limits of formal schemes in a wider context: the theorization of such limits is not peculiar to the Western analytic tradition. The note ends with an explicit statement of its own limitations (§7) and a provenance table recording the source of every substantive claim.

Throughout, we observe the following discipline: no citation is included without a locatable primary text or a verifiable secondary source; where a claim rests on secondary literature alone, this is stated; and no claim of the form "there is no evidence for X" is allowed to function as "there is evidence that not-X" (see §7).

---

2 Formalization: What Can and Cannot Be Encoded

2.1 The Language and the Encoding

We fix a first-order signature

```
L = { HCB( · ), CC( · ), J( · ), Kat( · ), Gr( · , · ), Stoic( · ) }
```

with the intended readings:

- `HCB(x)` —x is a Humean causal belief;
- `CC(x)` —x is produced by custom;
- `J(x)` —x is rationally justified (constrained by: $J(x) \to \neg HCB(x)$);
- `Kat(p)` —p is a kataleptic impression;
- `Gr(p, x)` —x is a cognition arising from the grasping of impression p;

- `Stoic(x)` —x is a Stoic-relevant belief.

The Stoic justification condition: a belief is justified only if it is a cognition arising from a kataleptic impression:

```
(Sto)  ∀z(Stoic(z) → ∃p(Kat(p) ∧ Gr(p, z)))
```

where `Stoic(z)` is a predicate for "z is a belief of the kind the Stoic theory is about". The Humean condition: a causal belief is produced by custom:

```
(Hum)  ∀z(HCB(z) → CC(z))
```

The exclusion claim under discussion: causal beliefs are not rationally justified:

```
(Exc1) ∀z(HCB(z) → ¬J(z))
```

or, formulated so that the exclusion flows *through* the mechanism of custom:

```
(Exc2) ∀z(CC(z) → ¬J(z))
```

Two observations are needed before we state the limit. First, the ancient sources define the kataleptic impression by a condition that is literally modal: the impression is one that "arises from what is, is stamped and impressed in accordance with what is, and is of such a kind as could not arise from what is not" (Diogenes Laertius 7.46 [Dorandi 2013]; Sextus Empiricus, Adversus Mathematicos 7.248 [Bury 1935]; 7.402 [Bury 1935]). The clause "of such a kind as could not arise from what is not" is an impossibility condition on the *source* of the impression: the veridicality of the impression is guaranteed by its causal provenance. Encoded in `L`, `Kat(p)` is an unanalysed predicate: the modal clause has been de-modalized by stipulation. Second, Hume's claim is not merely that causal beliefs are unjustified, but that they are unjustified *in virtue of being produced by custom* —the mechanism of belief formation excludes rational justification. This is an explanatory claim: the mode of production *grounds* the absence of justification (see §2.3).

2.2 What the Encoding Can Capture: The Material Content

It is important to be honest about what FOL *can* express here, because the limit we state in §2.3 is narrower than a careless reading of the literature suggests.

Let $T_1 = \{(\text{Sto}), (\text{Hum}), (\text{Exc1})\}$ and $T_2 = \{(\text{Sto}), (\text{Hum}), (\text{Exc2})\}$. Both theories are satisfiable: the finite model with domain $\{a\}$, Stoic $= \varnothing$, HCB $= \{a\}$, CC $= \{a\}$, Kat $=$ Gr $= \varnothing$, J $= \varnothing$ satisfies both (the Stoic axiom is vacuously satisfied). Moreover T_2 entails T_1: from (Exc2) and (Hum), by transitivity of the conditional, $\forall z(\text{HCB}(z) \to \neg \text{J}(z))$ follows. But T_1 does not entail T_2: in the model with domain $\{a, b\}$, HCB $= \{a\}$, CC $= \{a, b\}$, J $= \{b\}$, Stoic $=$ Kat $=$ Gr $= \varnothing$, the axioms of T_1 hold while (Exc2) fails at b ($\text{CC}(b) \wedge \text{J}(b)$). Hence the two formulations — "the exclusion attaches to causal beliefs directly" vs. "the exclusion flows through custom" —are extensionally distinguishable: they are different theories with different model classes. The material content of both readings is first-order expressible.

Nor is there any difficulty in representing the material co-occurrence facts. If one adds a predicate `E(x)` for "x is supported by adequate evidence" together with axioms relating `J` and `E`, the extension of `J` can be constrained as desired —but this is a *reduction*: it imposes a chosen reading of "justification" from outside the system, and the normative question ("what counts as adequate evidence, and why does custom fail to provide it?") remains external to the formalization. This is the classical point that first-order theories constrain the extension of their predicates but not their meaning; we sharpen it below.

2.3 What the Encoding Cannot Capture: The Explanatory and Modal Residue

The limit we state is semantic, not syntactic.

Model-pair argument. Consider any model $M \vDash T_2$ with a domain D of size $|D| \geq 1$, with an element $a \in D$ such that HCB(a), CC(a), and ¬J(a) hold in M. Two epistemologies are compatible with the *very same* L-structure M:

- *(E1)* ¬J(a) holds because CC(a) holds: the custom mechanism excludes rational justification; the belief's provenance explains its lack of justification.
- *(E2)* ¬J(a) holds "for no reason": the fact is stipulated, or explained by something unrelated to custom; the extension of `J` simply happens to exclude a.

Nothing in M distinguishes (E1) from (E2): first-order satisfaction is truth-functional, and the *explanatory ordering* between facts is not a component of a first-order structure. Since every L-structure admits both readings, the class of structures realizing (E1) and the class realizing (E2) coincide —both are the class of all L-structures. Consequently no first-order theory in the unexpanded signature can separate them, and *a fortiori* no consistent extension of T_2 with purely extensional axioms can force the intended normative reading of `J`. The distinction is not first-order definable because it is not a *structural* property at all. This is a semantic limit: the normative-explanatory content is not the sort of thing an L-structure can realize.

Two refinements are needed.

(a) The explanatory ("because") residue is hyperintensional. The claim "custom-production excludes justification" is naturally regimented as a grounding claim: for each causal belief z, the fact that CC(z) grounds the fact that ¬J(z) —in the notation of the contemporary grounding literature, CC(z) < ¬J(z) (Fine 2012; Correia & Schnieder 2012). Grounding is hyperintensional: it is not definable in terms of material implication, and not even in terms of strict implication (Fine 2012). The failure is therefore not specific to FOL's extensionality; it persists for modal extensions. This is why the draft's original suggestion — "a doxastic or modal operator suffices" —is *necessary but not sufficient*: a modal operator can express the *necessity* of the exclusion, but not the *direction of determination* between the mechanism and the absence of justification. The literature on "because" (Schnieder 2011) and on grounding agrees that the connective is non-truth-functional.

(b) The modal residue is de-modalized by the encoding. The Stoic definition's "could not arise from what is not" is a modal condition on the impression's provenance; the Humean claim, read strongly, is that the custom mechanism *cannot* yield rational justification —a necessity claim about a belief-forming process. Expressing either requires an accessibility relation over possibilities (a Kripke structure) —an *expansion of the ontology*, not a re-interpretation of L. It is true that modal formulas admit a standard translation into FOL over enriched structures (the "standard translation" of modal logic into first-order logic with a binary accessibility relation); but this translation changes the signature, and even then it captures *truth in a model*, not the hyperintensional "because". The relevant contrast for our argument is between the unexpanded language L and its possible enrichments: the intended reading is not in the expressive range of L.

Process-based epistemologies. There is a structural reason why both theories resist static encoding. Both Stoic *katalepsis* and Humean custom are *belief-forming processes*: the Stoic condition evaluates the reliability of the impression's source (the impression "could not arise from what is not"); the Humean condition describes a mechanism of association that is non-rational by constitution. Justification-by-reliable-process is exactly the subject matter of process reliabilism (Goldman 1979, 1–23; 1986): a belief is justified if it is produced by a reliable belief-forming process. A static, extensional language has no process dimension: it can record which beliefs were formed and which are justified, but not *the reliability of the process by which they were formed*—and it is precisely the process that carries the normative weight in both theories. This is the deep reason why the model-pair argument succeeds.

2.4 What a Richer Logic Adds: Justification, Grounding, Modality

The distinction that L cannot draw is representable in at least three richer frameworks. To keep the proposal testable, we give each option in the minimum formal shape in which it is expressive enough for the distinction.

(i) Justification logic. In justification logic (Artemov 2008; Artemov & Fitting 2019), formulas of the form $t{:}\varphi$ read "t is a justification (evidence) for φ", where t ranges over explicit terms. Formally, we enrich L to L^{JL} with two disjoint sorts of term: rational-ground terms $r_1, r_2, \cdots$ and custom-description terms $c_1, c_2, \cdots$, and add the rule of *application* $(s{:}(\varphi\to\psi) \wedge t{:}\varphi \Rightarrow (s \cdot t){:}\psi)$ with the factivity schema $t{:}\varphi \to \varphi$ (or, in weaker variants, without factivity). The Humean situation is then represented structurally: for a causal belief a, the theory contains $c{:}\mathrm{CC}(a)$—the custom term attests the production claim—while, at the meta-level, the theory has no theorem of the form $r{:}\mathrm{J}(a)$ for any closed rational-ground term r. The two roles that the extensional encoding collapses— "what produced the belief" and "what justifies it"—are separated at the level of the term language. This is the minimal positive moral of our negative result: the distinction requires a language in which *evidence* is an explicit first-class object.

(ii) Grounding. With a partial-grounding connective <, the exclusion is written $\mathrm{CC}(z) < \neg\mathrm{J}(z)$, with the standard axioms: irreflexivity $(\forall x{:}\ \neg(x < x))$, asymmetry $(\forall x\forall y{:}\ x < y \to \neg(y < x))$ and transitivity $(\forall x\forall y\forall z{:}\ x < y \wedge y < z \to x < z)$ (Fine 2012; Correia & Schnieder 2012). The direction of determination—the feature absent from model theory—is the very content of the connective. (These axioms are constraints on the intended reading; the expressivity claim does not depend on accepting the full logic of grounding.)

(iii) Modality. A modal operator over mechanism-relative possibility expresses the necessity claim: $\Box(\mathrm{CC}(z) \to \neg\mathrm{J}(z))$, i.e., the custom mechanism *cannot but* produce unjustified beliefs. As noted, this captures necessity but not direction; it is the appropriate vehicle if Hume's claim is read as a claim about the nature of the mechanism (see §3, point (c)).

We draw no conclusion about which framework is “the” correct one: the conclusion is disjunctive — *some* hyperintensional, modal, or explicit-evidence resource is required, and FOL is not such a resource at this granularity.

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3 Hume and the Principle of Sufficient Reason: A Qualification Required

A widespread reading says “Hume rejects the metaphysical PSR”; another says “Hume does not reject the PSR; he merely re-describes the question”. Neither is correct on its own. The verifiable facts are these.

(a) Hume denies that the causal maxim is rationally demonstrable. In *Treatise* 1.3.3, the maxim “whatever begins to exist must have a cause of existence” is said to be “neither intuitively nor demonstrably certain”: its contrary is conceivable, and our assent to it arises from observation and experience, i.e., from custom (T 1.3.3; SBN 78–82). The denial is therefore *epistemic*: the maxim is not rationally grounded.

(b) Hume explicitly denies absolute or metaphysical necessity in causal connection. T 1.3.14.35: “there is no absolute nor metaphysical necessity, that every beginning of existence shou’d be attended with such an object” (SBN 172).

(c) Hume’s positive account is psychological. T 1.3.14.20: necessity “is nothing but an internal impression of the mind, or a determination to carry our thoughts from one object to another” (SBN 164–65). The same thesis is stated in the *Enquiry*: “When we look about us towards external objects, and consider the operation of causes, we are never able, in a single instance, to discover any power or necessary connexion” (E 7.6; SBN 63). And the famous remark that “reason is nothing but a wonderful and unintelligible instinct in our souls” (T 1.3.16.9 SBN 178–9) makes the process-based character of the account explicit: belief formation is a mechanism, not an inference.

Hence: Hume does not “refute” the PSR in its causal form, and he does not leave it untouched; he removes it from the level of demonstration and relocates it at the level of custom. The reading “Hume denies the PSR” is false if it means he argued against the principle’s *content*; the reading “Hume never questions it” is false if it means he accepted it as a rational principle. The correct formulation is: the causal maxim is, for Hume, a natural psychological regularity masquerading as a rational necessity.

Two further points. First, the term “Principle of Sufficient Reason” (*principium rationis sufficientis*) is Leibnizian (*Monadologie* §32; 1714); Hume never uses it, and retro-application requires care. Second, the literature on Hume and the causal maxim is substantial: on the maxim itself, see Garrett 1997, 24, 66–69; on the projectivist reading of causal necessity, Beebee 2006; on the *Enquiry* context, Millican 2002, 27–65. The contemporary revival of PSR debates (Della Rocca 2010; Pruss 2006) has made the question of what exactly Hume denied once more a live interpretive issue; our claim (a)–(c) is deliberately minimal and textually anchored.

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4 The Transmission of Stoic Epistemology to Early Modern Europe: A Verifiable Route

4.1 Sextus Empiricus

The kataleptic impression and the criterion debate are documented most fully in Sextus Empiricus, *Adversus Mathematicos* VII: the criterion discussion at M 7.24–29, the definitional texts at M 7.248 and 7.402, and the "clear, evident, striking" (enarges, ektypos, plektike) marks at M 7.257–58 (cf. Bury 1935, 7.402–8). The printing history is now well documented (Floridi 2002; Popkin 1979, 18): the first Latin translation of the *Hypotyposes* was published in 1562 by Henri Estienne (Geneva); the Latin *Adversus Mathematicos*, together with a second edition of the Estienne translation, was published in 1569 by Gentian Hervet (Paris: Martin Juvenis; Antwerp: Plantin). The Greek editio princeps of Sextus' *Opera* appeared only in 1621 (Geneva: Chouët). The common claim that "Hervet translated Sextus in 1562 and 1569" therefore conflates Estienne with Hervet and should be corrected.

4.2 Diogenes Laertius

The main doxographical source for Zeno's and Chrysippus' epistemology (the definitions of *phantasia*, *phantasia kataleptike*, *katalepsis*, *epistēmē* at *Vitae Philosophorum* 7.46–54; the criterion at 7.54 [Dorandi 2013]; cf. Bobzien 2003, 85–123) reached early modern Europe through Ambrogio Traversari's Latin translation, completed in manuscript by 1433; the Latin editio princeps was printed in Rome 1472 (Georgius Lauer); the celebrated Venice 1475 (Nicolaus Jenson) is the second printed edition; the Greek editio princeps appeared in Basel 1533 (Froben) (Dorandi 2013, 15; ISTC id00219000/id00220000).

4.3 Cicero and the Neostoic Reception

Cicero's *Academica* is the Latin source for Zeno's fist analogy—open hand: impression; fingers bent: assent; fist: *katalepsis*; the other hand grasping the fist: *epistēmē*, available to the sage alone (*Lucullus* 145 [Brittain 2006]; SVF 1.66)—and for the criterion debate in Latin. The *Academica* entered print early: editio princeps Rome 1471 (Sweynheym & Pannartz), with the Aldine edition of 1523 establishing the standard text (Hunt 1998). But the decisive early modern channel for Stoic doctrine as a *system* was Neostoicism: Justus Lipsius published *De Constantia* (Antwerp: Plantin, 1584), and in 1604, both the *Manuductio ad Stoicam philosophiam* and the *Physiologia Stoicorum* (Antwerp: Plantin-Moretus). The reception of Stoic ethics through Lipsius and du Vair is well documented (Lagrée 1994; du Vair 1594); what matters for this note is that the *epistemological* transmission—the katalepsis doctrine—had to proceed through Sextus, Diogenes Laertius and Cicero, since the Neostoics' interest was primarily ethical.

4.4 Hume's Own Access

For Hume's own acquaintance with these materials, the documented facts are these. The catalogue of Hume's library (Norton & Norton 1996) lists Cicero's *Opera* (Paris, 1740–42, 9 vols.); it does not list Sextus Empiricus. The 1840 sale catalogue, on which the reconstruction rests, shows no Sextus (Fosl 1998). However, the 1840 catalogue reflects books sold at auction; it is not identical to the full library at Hume's death, as books were often given away or sold earlier. The absence of Sextus is therefore negative evidence, not a proof of Hume's ignorance. But building a "Hume owned a Sextus" narrative on this gap is historiographically irresponsible. The standard scholarly account remains: Hume's access to Pyrrhonism ran through Bayle, Montaigne and others (Popkin 1952, 65–81; 1951). The historiography of "Hume the Pyrrhonian" is best grounded in: (i) Hume's direct reading of Cicero; (ii) his mediated reading of Sextus through the French sceptical tradition.

4.5 The Medieval Question and the Cartesian Comparison

Linear chains of the form "Stoicism → Aristotelian commentators → Scholasticism → Descartes → Locke" are not documented. The evidence for discontinuity is strong: only seven Latin Sextus manuscripts circulated in the medieval Latin West (Floridi 2002); the *Academica* was among the rarer Cicero texts in the Middle Ages, with Henry of Ghent the first scholastic explicitly to engage it (Schmitt 1972; 1983; SEP "Medieval Skepticism"); the term "scepticus" entered Latin usage only in the 1430s. Two qualifications are required, in the spirit of §7. First, "no continuous direct transmission of the *doctrine of katalepsis*" does not mean "no knowledge of Stoic epistemology at all": Cicero's *comprehensio* and Augustine's *Contra Academicos* provided an indirect, contested channel (Nawar 2022, 185–207). Second, the relation between Descartes's "clear and distinct ideas" and the kataleptic impression should be studied as *discrete responses to a common skeptical problem* —the criterion problem —rather than as a transmission chain; this is the position of the relevant secondary literature (Frede 1983; Nawar 2022).

Similarly, Locke's critique of innate principles in *Essay* I.3 ("No Innate Practical Principles") is directed, in its explicit form, at Herbert of Cherbury's *De Veritate* (1624): Herbert is named at I.3.15 and discussed through I.3.27 (Nidditch 77–84); Descartes is not the primary target of that chapter. And the very notion of a unified "British Empiricism" is a construction of nineteenth-century historiography (Norton 1981). The moral of this section: the history of philosophy should be written from the history of printed texts, not from speculative chains.

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5 A Comparative Note: Formal Limits Across Philosophical Traditions

The theme of this note —that a formal scheme has limits that must be stated from within and beyond —is not peculiar to the Western analytic tradition. We record three documented parallels and note two further items for completeness. Each is invoked as an analogy only: no claim of historical transmission between these traditions and the Stoic/Humean material is made or implied (cf. §7).

(a) The *catuṣkoṭi*. The Buddhist four-cornered negation schema —*P*, ¬*P*, *both*, *neither* —was theorized by Nāgārjuna as an exhaustive decomposition of assertoric space, and it has been shown that classical two-valued semantics cannot express the fourth corner as a distinct position; the schema is naturally formalized in paraconsistent logics (Priest 2010, 24–54; 2018). One asymmetry must be stated honestly: the *catuṣkoṭi* is a case of *expanding* a formal scheme —paraconsistency extends the logic — whereas our §2 result is a case of a scheme's *representational limit*. The parallel is therefore not in the kind of limit but in the practice: the *catuṣkoṭi* shows that theorizing about what a scheme can and cannot say —including the scheme's own boundary —is a distinguished non-Western philosophical activity. Whether Nāgārjuna himself intended the *catuṣkoṭi* as a meta-logical reflection on the limits of assertoric space, or whether this is a modern reconstruction (Priest 2018, ch. 1; Tillemans 1999), is itself a contested interpretive question. The claim is not that Stoic or Humean materials require paraconsistency; whatever Nāgārjuna's intentions, the *catuṣkoṭi* schema itself —as an exhaustive decomposition of assertoric space —encodes a boundary-theorizing practice, and so the question "what cannot be represented here?" is not a modern Western invention.

(b) The Xunzian rectification of names (*zhengming*). Xunzi's chapter "On the Rectification of Names" (Xunzi 22) addresses the criterion problem for language: names are correct if fixed by shared sensory experience (the *tianguan*, "heavenly faculties", common to all humans —Xunzi 22.5) and by convention ("names have no intrinsic appropriateness; they are fixed by agreement" —Xunzi 22.9; Hansen 1992, ch. 9; Hansen 1983). The structural parallel with the Stoic criterion debate is striking and has been noted in the comparative literature (Rošker, SEP "Epistemology in Chinese Philosophy"; Kjellberg 1996): both theories face the question of what fixes the *standard* by which representations are judged correct. The differences are equally instructive. At the level of the *kind of standard*, the Stoic criterion is source-based (the impression's provenance) while the Xunzian criterion is intersubjective-conventional. But the deeper difference is in the *functional position* each doctrine occupies within its system. The Stoic *katalepsis* is an epistemological doctrine about individual cognitive accuracy: it serves to secure truth for a single cognitive agent, and its failure mode is epistemic error. The Xunzian *zhengming*, by contrast, is an ethical-political doctrine about the ordering of social life: its purpose is to ensure that correct naming yields correct action and social coordination (cf. Xunzi 22.9; the canonical formula "if names are not correct, language will not be in accordance with truth" is Analects 13.3 —the Xunzian claim is a development in the same spirit rather than a verbatim citation), and its failure mode is social disorder. The structural parallel at the level of "what fixes the standard?" thus masks a difference in the *domain* and the *end* to which the standard is put: the analogy should not be read as implying that the two doctrines occupy the same functional position in their respective systems. A first-order encoding of either faces the same §2.3 problem: the criterion's *normative force* —in virtue of what the standard is the standard —is not a truth-condition.

(c) *Kaozheng* and documentary discipline. The Qing-dynasty *kaozheng* (考據, "evidential research") movement —exemplified by Dai Zhen (*Mengzi ziyi shuzheng*, Preface; in *Dai Zhen wenji*, Zhonghua Shuju, 1980; cf. Elman 1984, Introduction and ch. 5; 2nd ed. 2001) —made textual verification, philological evidence and the rejection of unfounded commentary the methodological core of scholarship. Our §4 and §7 defend precisely this discipline for the history of philosophy: the claim that documentary discipline is a method, not an ornament, has a systematic analogue in a non-Western tradition that is still under-cited in analytic historiography. The parallel is in the *methodological attitude* —evidence over speculation —not in the object of inquiry, which differs. Again: analogy, not influence.

We also note, for completeness, the Mohist tradition' s treatment of the name–reality (名/實) distinction and the argumentation chapter "Lesser Selection" (Xiaoqu) of the *Mozi* (Graham 1978; 1989), and the White Horse paradox of Gongsun Long, which turns on the gap between a predicate' s linguistic ontology and its extension —a theme in the vicinity of, though not identical with, the model-theoretic underdetermination discussed in §2.1 (Hansen 1983, ch. 5). Finally, Wang Yangming' s doctrine of *liangzhi* ("innate knowing" ; *Chuanxilu* 2.2; Van Norden, SEP "Wang Yangming") treats the cognitive standard as an inner, quasi-evidential faculty rather than an external impression or convention —structurally closer to the Stoic katalepsis than to the Xunzian convention, since both posit a source-based standard rather than an intersubjective one, yet differing in that Wang Yangming' s doctrine is embedded in a moral-spiritual project (the unity of knowledge and action) rather than primarily in an epistemological criterion debate. [Provenance note: the chapter and passage numbers above follow the standard editions cited in the bibliography; where translations differ (Watson; Knoblock), we rely on the philological consensus reported in Hansen 1992 and the Stanford Encyclopedia entries cited. The claim that the Stoic/Xunzian parallel has been noted in the comparative literature is supported by Kjellberg 1996 (Sextus/Zhuangzi/Xunzi on skepticism) and by the SEP entry "Epistemology in Chinese Philosophy" (Rošker); both are cited as reporting a parallel, not as establishing a historical connection.]

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6 Conclusion

This note has made three contributions.

1. Technical limit. A purely extensional first-order encoding of the justification conditions of Stoic *katalepsis* and Humean custom can represent the material content of both positions —including the extensionally distinguishable formulations (Exc1) and (Exc2) —but cannot represent their explanatory and modal residue: the "because" of the Humean exclusion (hyperintensional; grounding), the "could not arise from what is not" of the Stoic definition (modal), and the reliability of belief-forming processes. The model-pair argument shows the limit is semantic: the intended reading is not a structural property of L-models. A justification operator (Artemov & Fitting 2019), a grounding connective (Fine 2012), or a modal operator is required. This is a singular encoding limit, not a universal failure claim.

2. Interpretive limit. Hume' s relation to the PSR is neither simple rejection nor silence: he relocates the causal maxim from the demonstrative level to the level of custom (T 1.3.3; SBN 78–82; T 1.3.14.20; SBN 164–65; T 1.3.14.35; SBN 172). "Hume never questions the PSR" and "Hume refutes the PSR" are both false; the accurate formulation is a *demotion*, not a refutation. A significant interpretive counter-thesis —defended notably by Della Rocca (2010) and, in a different register, by Pruss (2006) —argues that this "demotion" is a polite fiction: by reducing the PSR to a

"natural psychological regularity," Hume effectively *evaporates* its normative status, rendering the distinction between "demotion" and "rejection" functionally moot. We maintain the demotion formulation as the more textually anchored reading, since Hume explicitly preserves the causal maxim as a psychological principle rather than arguing against its content; but the evaporation reading remains a strong interpretive alternative, and our claim should be read as a *preference* supported by the textual evidence, not as a settled result.

3. Historical limit. The transmission of Stoic epistemology to early modern Europe runs through documented printed texts: Sextus (Latin 1562 Estienne; 1569 Hervet; Greek 1621), Diogenes Laertius (Traversari 1433; Latin editio princeps Rome 1472; Venice 1475; Greek Basel 1533), Cicero's *Academica* (in *Opera omnia*, Rome 1471 [Sweynheym & Pannartz]; Aldine edition 1523; the fist analogy at *Lucullus* 145 [Brittain 2006]), and the Neostoic reception (Lipsius 1604). Speculative medieval chains are not documented; indirect medieval knowledge via Cicero and Augustine is. Hume's own access ran through Cicero directly and Sextus through Bayle (Popkin 1952, 65–81).

Formal tools are useful precisely when their limits are stated; the history of philosophy requires a documentary discipline that formal notation cannot supply; and both theses have systematic, independently developed analogues in the Buddhist, Xunzian and kaozheng traditions (§5).

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7 Limits of This Note

In the same spirit, the limits of this note are stated explicitly.

1. We make no claim of *in-principle* inexpressibility for any language richer than L: the negative result is about this signature, at this granularity, under extensional semantics.
2. We do not claim that justification logic, grounding, or modal logic "solves" the philosophical problems; we claim only that they make the relevant distinction representable.
3. The comparative section (§5) is analogical; none of its claims depends on a historical connection between the traditions compared, and none asserts one.
4. All historical claims rest on the printed-text history summarized in §4; where the secondary literature is the sole support (e.g., the medieval manuscript census), this is stated, and the sources are given.
5. "No evidence of continuous direct transmission" (§4.5) is a negative-evidence claim; it is recorded as such and is not asserted to be a proof of non-transmission.
6. Quotations follow the editions listed under "Citations and Editions"; paragraph numbers for Hume follow the Norton & Norton (2000) and Beauchamp (1999) editions, with Selby-Bigge/Nidditch (SBN) pagination given in every case.
7. The formal encoding in §2.1 is itself an interpretive choice. Different readings of the Stoic and Humean sources might yield different signatures or axioms; our result is relative to the encoding chosen, and the choice of encoding is a hermeneutic decision that precedes the formal argument.

8. The richer frameworks of §2.4 —grounding, justification logic, process reliabilism —are invoked as *contemporary regimentations* of the relevant claims; no claim is made that Hume or the Stoics held these theories, and their vocabulary is applied in a deliberately anachronistic, analytic sense.

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8 Appendix: Definitions

Term	Definition	Source
katalepsis	Stoic "cognition": the grasping of a kataleptic impression; the criterion of truth for the Stoics	DL 7.46, 7.54 [Dorandi 2013]; translation "cognition": Long & Sedley 1987, 40; SEP "Stoicism" §3.7
phantasia kataleptike	Kataleptic impression: "arising from what is, stamped and impressed in accordance with what is, of such a kind as could not arise from what is not"	DL 7.46 [Dorandi 2013]; M 7.248 [Bury 1935]; 7.402 [Bury 1935]
Criterion problem	The question what standard fixes which impressions are true; the regress of "criterion of the criterion"	Sextus M 7.24–29; PH I.164–169 (169 = diallelos); M 7.440–446
Assent (*sunkatathesis*)	The Stoic act of approving an impression; the middle stage between impression and cognition	DL 7.46–54; Cicero, *Acad.* 2.37–39, 2.66–67
PSR	*Principium rationis sufficientis*: Leibnizian term; every fact has a sufficient reason	Leibniz, *Monadologie* §32 (1714)
FOL	First-order logic: extensional, truth-functional model-theoretic semantics	Standard; §2
Justification logic	Logic with formulas $t{:}\varphi$, "t is a justification for φ", for explicit terms *t*	Artemov & Fitting 2019
Grounding	Hyperintensional explanatory relation: $\varphi < \psi$, "φ grounds ψ"	Fine 2012; Correia & Schnieder 2012
Custom	Humean habit: determination of the mind to pass from one object to its usual attendant	T 1.3.14.20; E 5
catuṣkoṭi	Buddhist four-cornered negation: P, ¬P, both, neither	Priest 2010

Term	Definition	Source
zhengming (正名)	Xunzian rectification of names: fixing correct usage by sensory agreement and convention	Xunzi 22.5, 22.9; Hansen 1992
kaozheng (考據)	Qing evidential scholarship: philological verification as method	Elman 1984, Introduction and ch. 5

Citations and Editions

- Hume, *Treatise* (T): Norton & Norton (eds.), Oxford University Press, 2000; paragraphs cited as T 1.3.3 etc.; Selby-Bigge/Nidditch (SBN) pagination in parentheses.
- Hume, *Enquiry* (E): Beauchamp (ed.), Oxford University Press, 1999; paragraphs cited as E 7.6 etc.; SBN pagination in parentheses.
- Sextus Empiricus: *Primary source editions*: Latin translations —*Hypotyposes*, tr. Hervet, publ. Estienne, 1562; *Adversus Mathematicos*, tr. Hervet, 1569; Greek *Opera*, Chouët, 1621. *Modern translation used*: Bury (tr.), Loeb Classical Library, 1933–49.
- Diogenes Laertius: Hicks (tr.), Loeb Classical Library, 1925; Dorandi (ed.), Cambridge University Press, 2013.
- Cicero, *Academica*: Brittain (tr.), Hackett, 2006; *Lucullus* paragraph numbers (Plasberg).
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- von Arnim, H. (ed.). *Stoicorum Veterum Fragmenta* (SVF), vol. 1. Leipzig: Teubner, 1905.

Provenance Table

§	Claim	Source
Abstract	katalepsis = “cognition”(Long & Sedley 1987, 40)	Long & Sedley 1987, 40; cf. SEP “Stoicism” §3.7 [P-01]
§2.1	Kataleptic definition with modal clause	DL 7.46 [Dorandi 2013]; M 7.248 [Bury 1935]; 7.402 [Bury 1935] [P-03]
§2.2	$T_1 \not\models T_2$, model exhibited	Formal argument, this note (model: {a,b}; HCB={a}; CC={a,b}; J={b}) [P-04]
§2.3	Grounding hyperintensional	Fine 2012; Correia & Schnieder 2012 [P-05]

§	Claim	Source
§2.3	Process reliabilism	Goldman 1979, 1–23; 1986 [P-05]
§2.4	Justification logic	Artemov 2008; Artemov & Fitting 2019 [P-06]
§3	T 1.3.3; SBN 78–82	Hume, T (Norton & Norton 2000); SBN [P-07]
§3	T 1.3.14.20; SBN 164–65	Hume, T [P-07]
§3	T 1.3.14.35; SBN 172	Hume, T [P-07]
§3	E 7.6; SBN 63	Hume, E (Beauchamp 1999) [P-07]
§3	T 1.3.16.9; SBN 178–9	Hume, T [P-07]
§3	PSR literature	Garrett 1997; Beebee 2006; Millican 2002; Della Rocca 2010; Pruss 2006 [P-07]
§4.1	1562 Estienne; 1569 Hervet; Greek 1621	Floridi 2002; Popkin 1979, 18 [P-08]
§4.1	M 7.24–29; M 7.257–58	Sextus, M (Bury, Loeb) [P-08]
§4.2	Traversari 1433; Rome 1472; Venice 1475; Basel 1533	Dorandi 2013, 15; ISTC id00219000/id00220000 [P-09]
§4.3	Fist analogy at Lucullus 145 [Brittain 2006]	Cicero, Acad. (Brittain 2006); SVF 1.66 [P-10]
Abstract	Fist analogy = Long & Sedley 41B	Long & Sedley 1987, section 41B (= Cicero, *Lucullus* 145) [P-01]
§4.3	Lipsius 1584; 1604	SEP “Justus Lipsius”; Lagrée 1994 [P-10]
§4.4	Hume’s library: Cicero yes, Sextus no	Norton & Norton 1996; Fosl 1998 [P-11]
§4.4	Bayle-mediated access	Popkin 1952, 65–81; 1951 [P-11]
§4.5	Medieval discontinuity; nuances	Floridi 2002; Schmitt 1972; SEP “Medieval Skepticism”; Nawar 2022 [P-12]
§4.5	Locke–Herbert I.3.15–27	Locke, Essay (Nidditch 1975), 77–84 [P-12]
§4.5	British Empiricism myth	Norton 1981 [P-12]
§5	Catuṣkoṭi formalized	Priest 2010; 2018 [P-13]
§5	Xunzi 22.5; 22.9	Xunzi (ctext; Knoblock tr.); Hansen 1992, ch. 9 [P-13]
§5	Kaozheng	Elman 1984, Introduction and ch. 5; 2nd ed. 2001 [P-13]
§5	Mohist/Gongsun Long	Graham 1978; 1989; Hansen 1983, 140–51 [P-13]
§7	Limits stated	This note [P-15]

SC-ID mapping (CHK-02). Every mandatory SC-ID corresponds to at least one section covered by the table: SC-001–004 → Abstract/Özet/Keywords; SC-005 → §1; SC-006–012 → §2.1–2.4; SC-013–014 → §3; SC-015–017 → §4; SC-018 → §5; SC-019 → §6; SC-020–022 → Provenance/Citations/References;

SC-023 → Appendix: Definitions; SC-024–030 → §4.4, §4.5, Appendix, §7 (optional items are covered where adopted).

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